

STRUCTURAL DESIGN SESSIONAL FINAL REPORT TEAM - 5



Course Instructors:

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Table of Content:

1. Group Members
2. Design Basis Report
3. General Arrangement Drawing and Building Design Plans
4. Isometric, Front & Side View of Tower
5. Beam Column Layout
6. Load Combinations & Reaction Forces
7. Analysis of the Structure
8. Load Calculations
9. Design Calculations
10. Design of Shear Walls & Foundation
11. Structural Detailing of Beam-Column Junction
12. Plumbing Network Analysis
13. Road Network Analysis
14. Cost Estimation

Group Members:

Loghith US (23CE10029)

Keerthi Modalavalasa (23CE10032)

Lovekush Meena (23CE10030)

Krupal Thakre (23CE10028)

Debmalya Kar (23CE30012)

Divyajot Bhalla (23CE30013)

Design Basis Report

Problem Statement

Design a steel concrete composite residential tower comprising five apartments in each floor with the following information supplied.

Use the following information:

1. The campus is located at Chandigarh for Odd group numbers and at Bhubaneswar for even group numbers.
2. Each Residential Tower will be of $(B+G+5+2*\text{Group Number})$ storey for group No. I to VIII; $B+G+2*\text{Group Number}$, for group No. IX to XV.
3. Each floor will have five apartments; three studio apartments (built-up area = 35 ± 10 m² each) and two 2-BHK apartments (built-up area = 55 ± 10 m² each) with common facilities.
4. The bearing capacity of the soil for towers with odd group number is 12 ton/m² while that for towers with even group number is 18 ton/m².
5. Clear height of the, basement and ground floor is 2.50 m
6. The height of each floor is 3.0 m for group number I to V; 3.2 m for group number VI to X; 3.5 m for group number XI to XV.
7. Analysis and design of the Towers should comply with the latest Indian Codes of Practice.

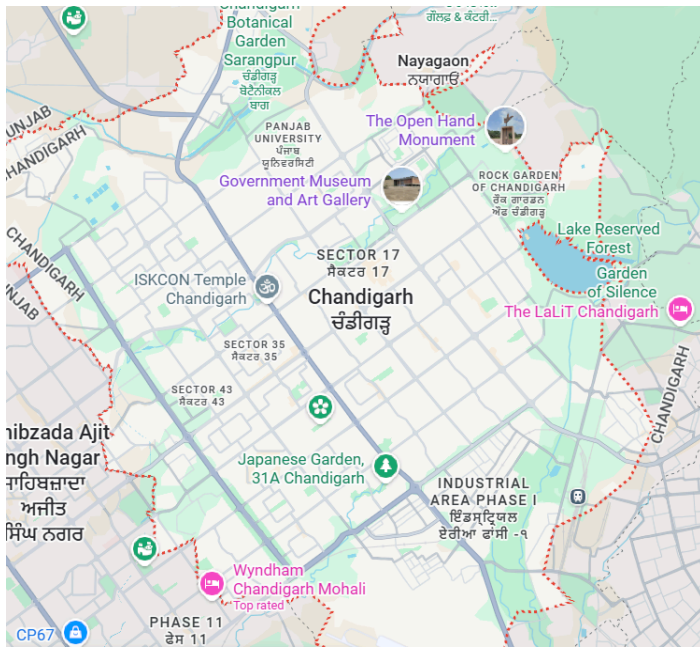
Material Considered:

- For all RCC structural elements, M30 grade of concrete and Fe500 steel will be used.
- For steel members: Fe410 grade will be used.

General Conditions:

9. The floor diaphragms are assumed to be rigid.
10. Centre-line dimensions will be followed for analysis.
11. Seismic loads will be considered acting in the horizontal direction and not along the vertical direction.
12. Any other parameter needed for the design can suitably be assumed with proper justification.

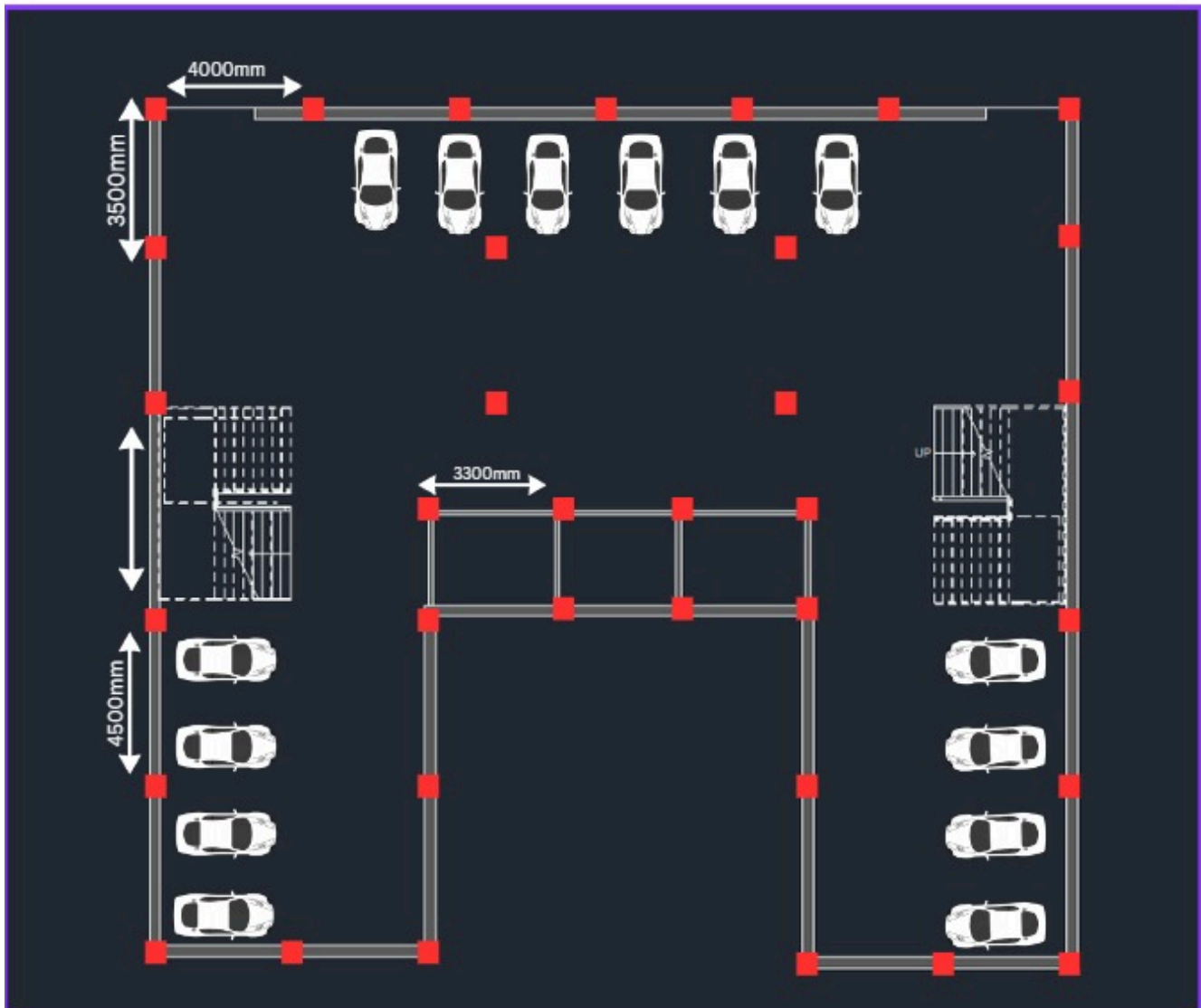
Tower Plan & Design:



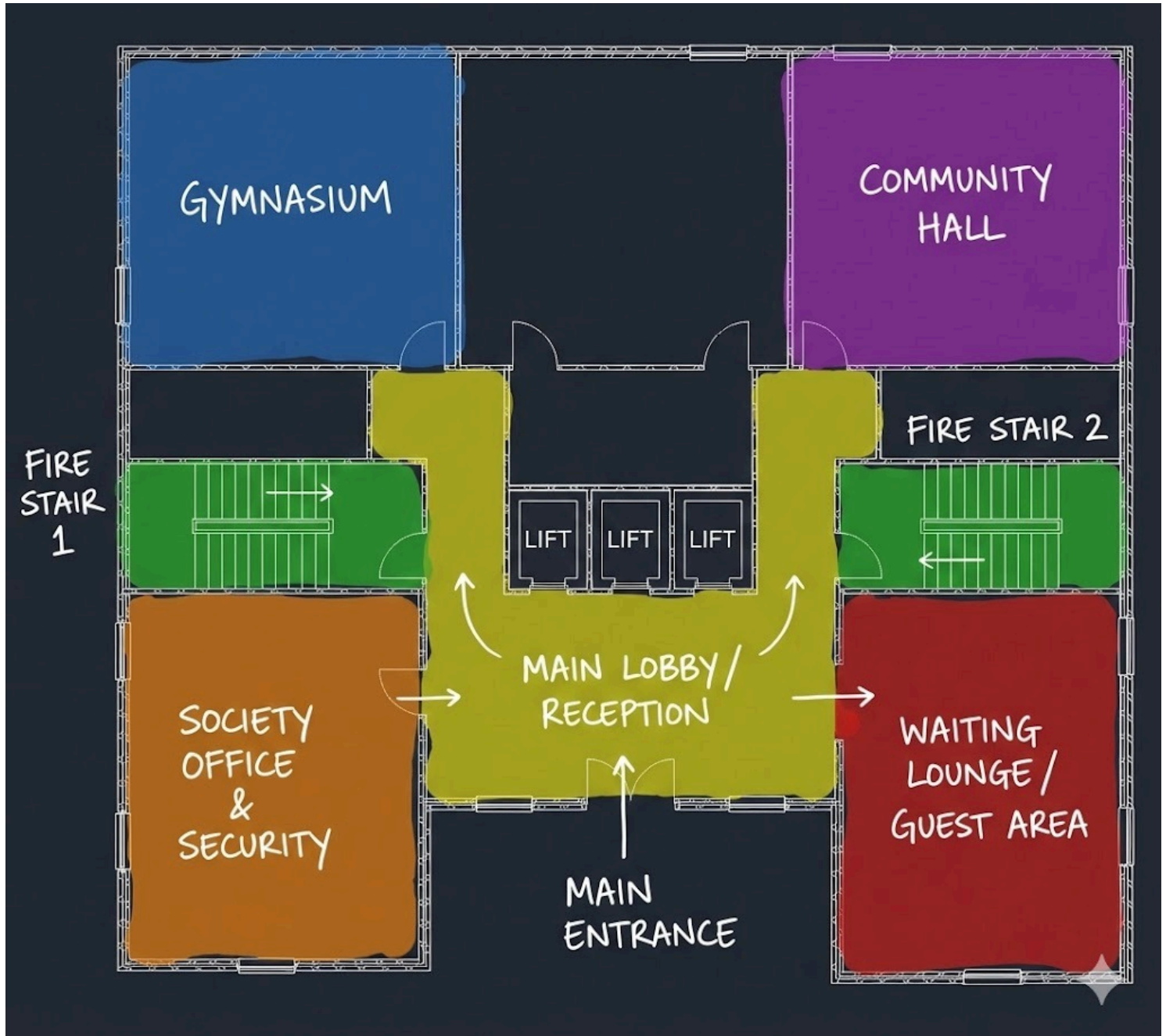
- Location - Chandigarh
- Seismic zone - Zone 4
- Wind Zone - Moderate to High wind speed zone (Basic wind speed = 47 m/s)
- Total no. of floors - B+G+15 floors
- Basement floor - 1 floor (2.5 m)
- Ground floor - 2.5 m
- No. of residential floors - 15 (3 m)

- This residential tower has 5 apartments on each floor. Since there are 15 floors, a total of 75 apartments are present in this complex.
- The two 2-BHK apartments have similar dimensions and plan, and the three Studio apartments have similar plans.

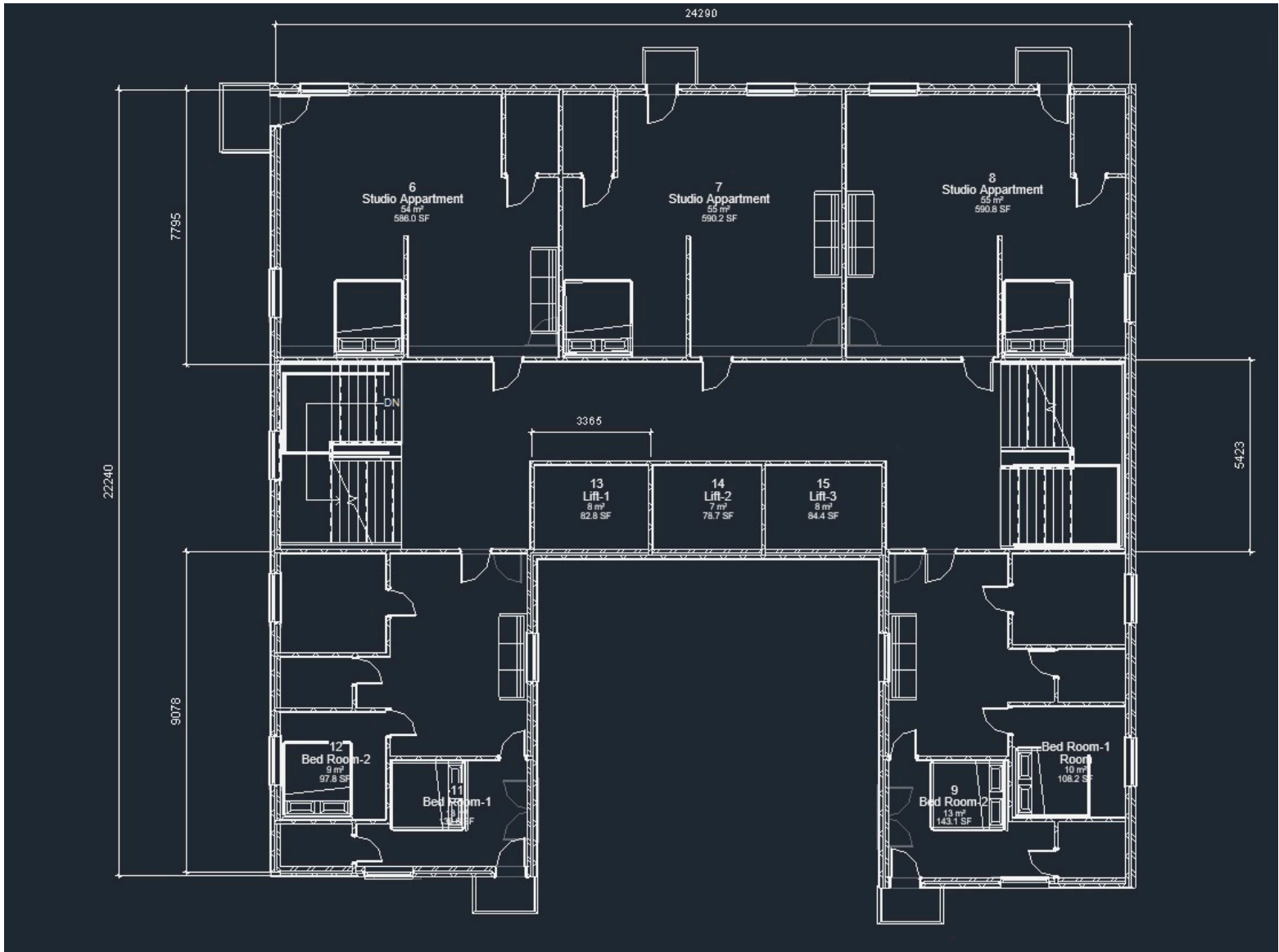
Basement plan:



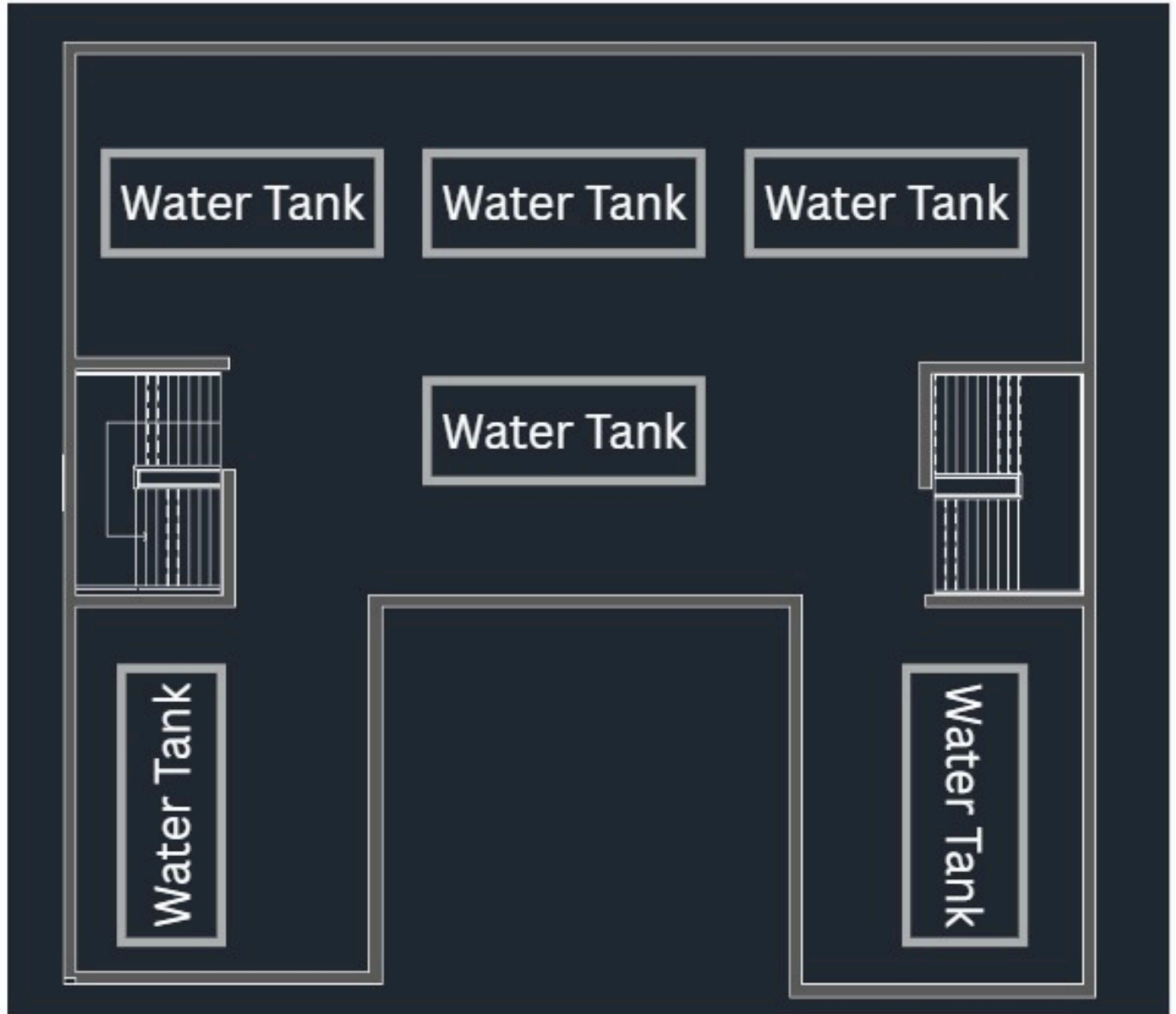
Ground floor plan:



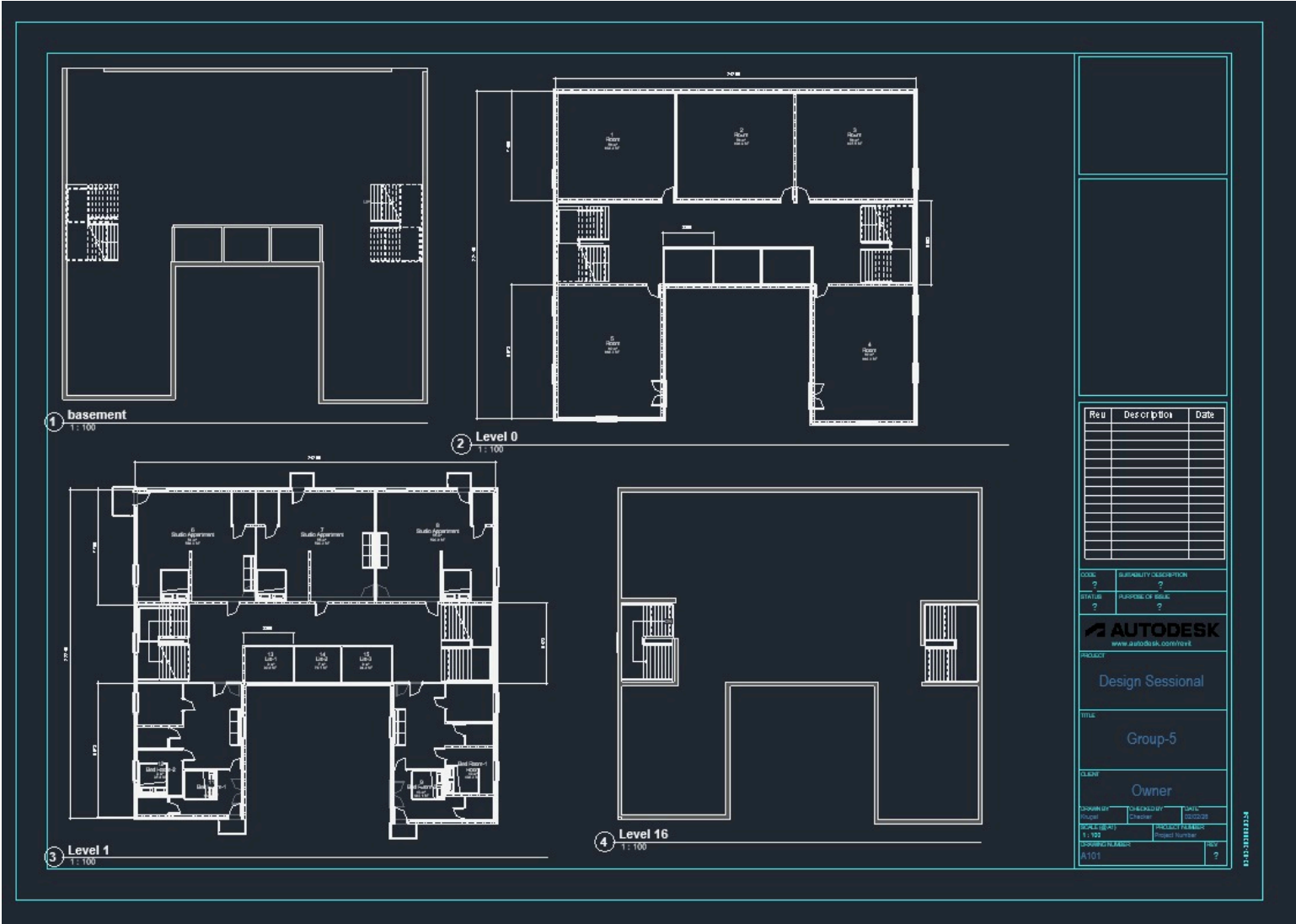
Typical Floor plan:



Roof Plan:

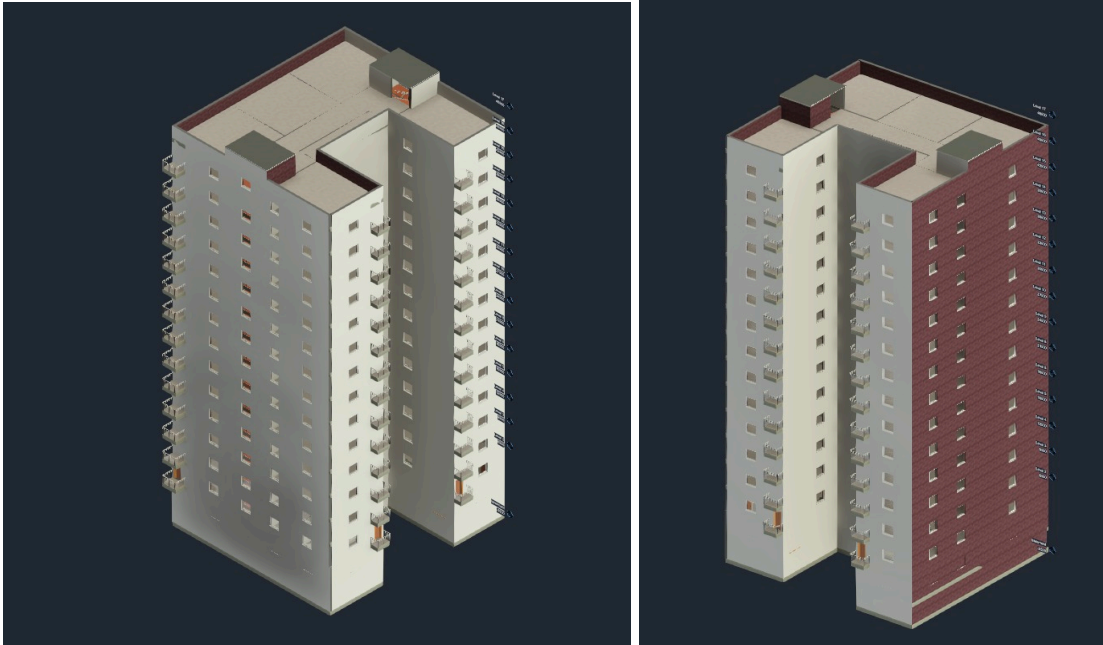


General Arrangement Drawing:

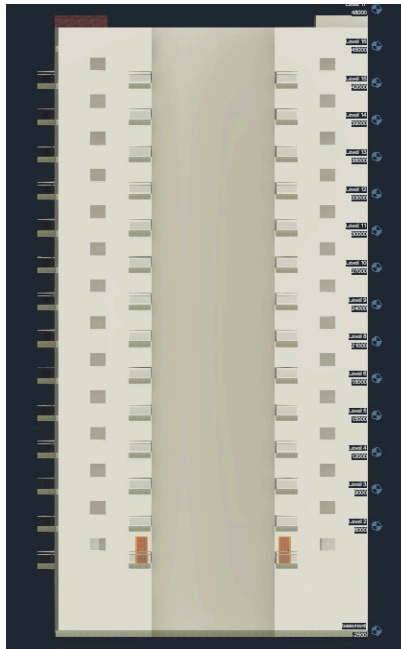


Isometric, Front and Side views:

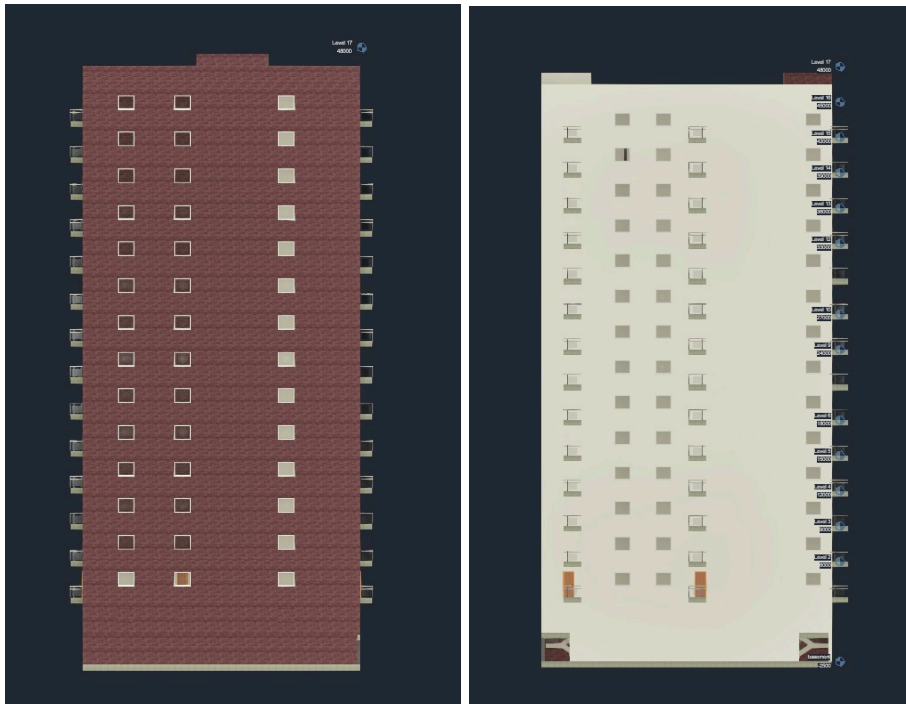
Isometric View:



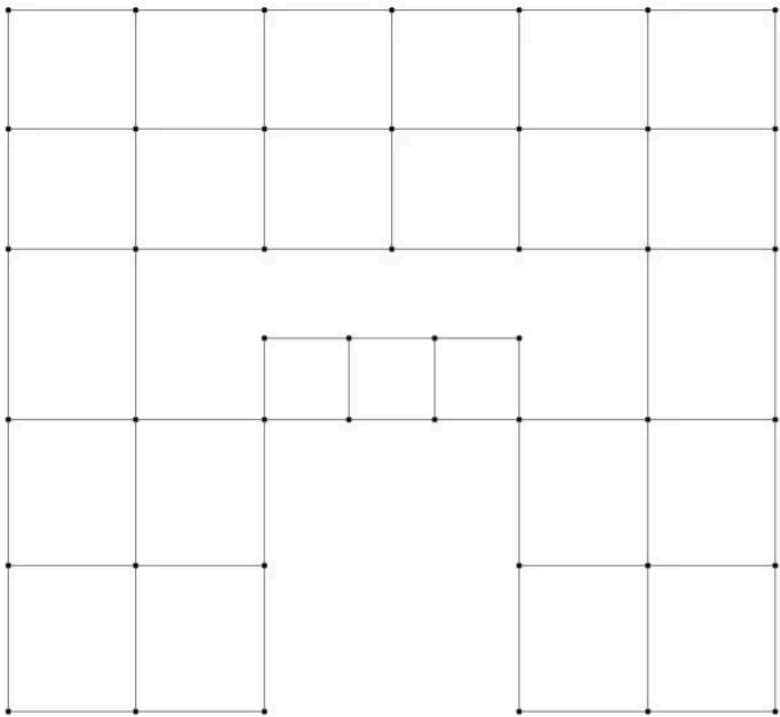
Front View:



Side View:

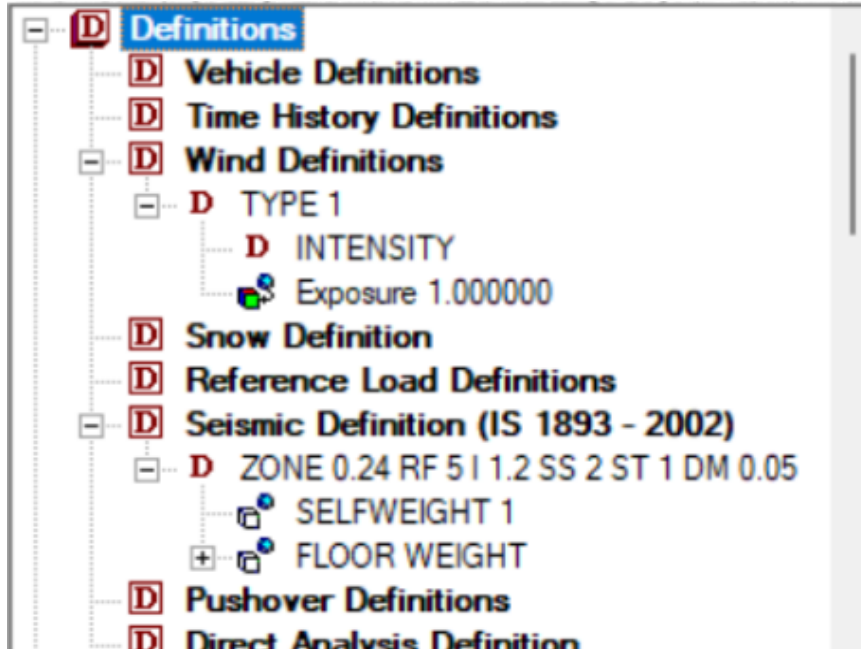


Beam Column Layout:



Load Combinations:

Load Definitions:



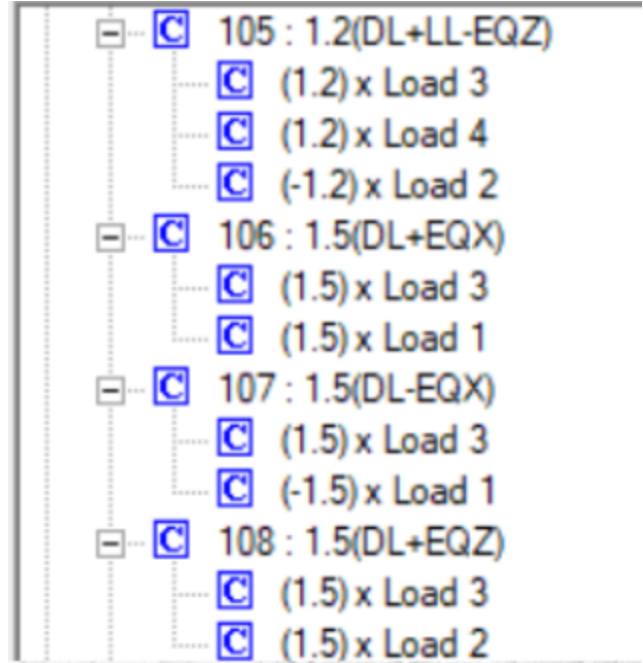
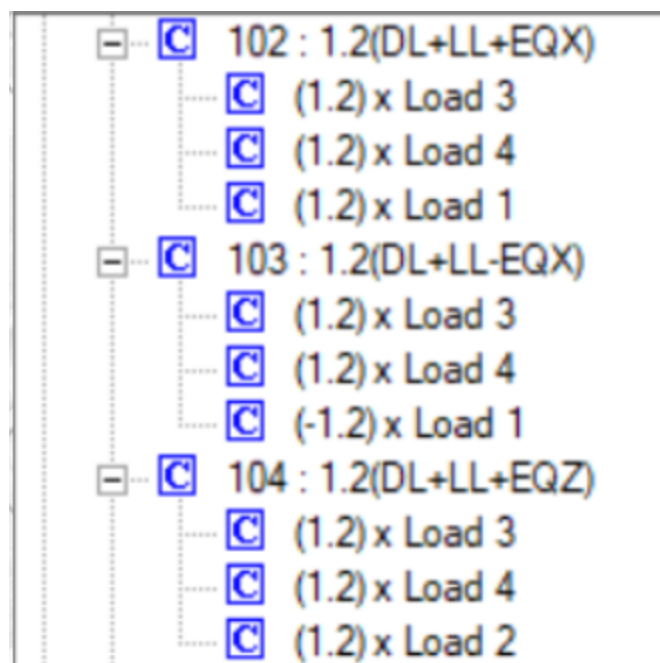
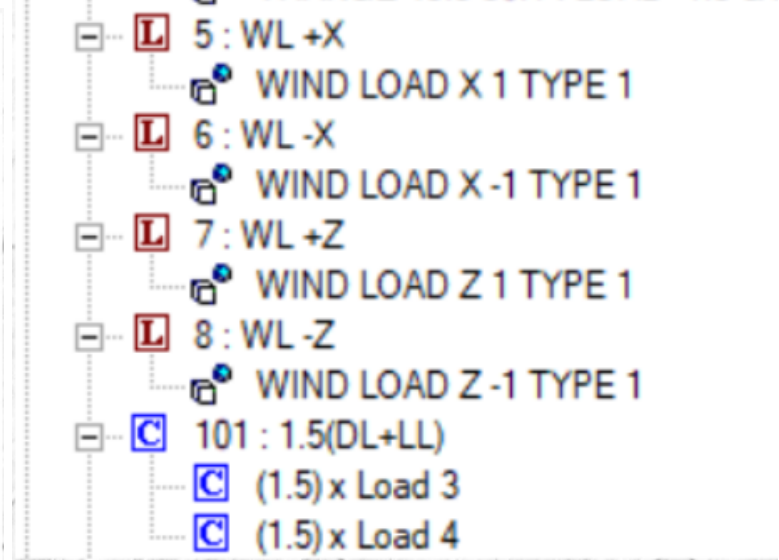
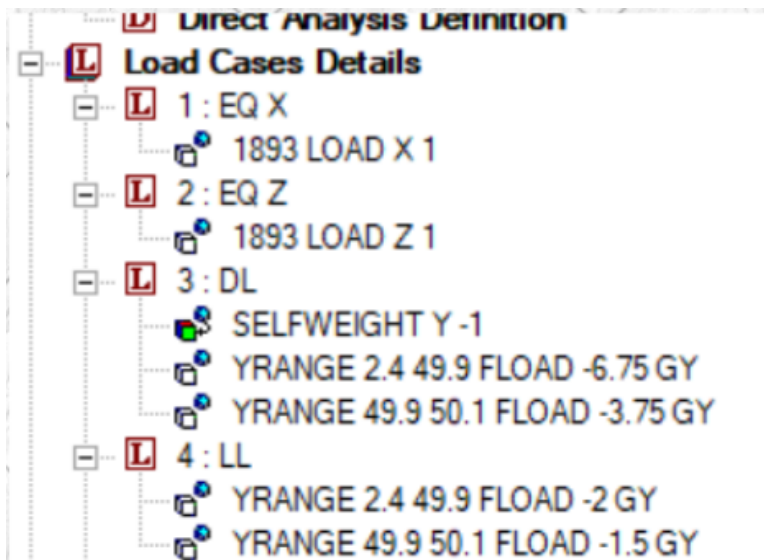
Intensity

Select Type: Custom

Intensity vs. Height

	Int (kN/m ²)	Height (m)
1	1.090000033	10
2	1.240000009	15
3	1.350000023	20
4	1.480000019	30
5	1.720000028	50
6		

Load Case Details & Load Combinations:



-	☐	109 : 1.5(DL-EQZ)
	☐	(1.5) x Load 3
	☐	(-1.5) x Load 2
-	☐	110 : 0.9DL+1.5EQX
	☐	(0.9) x Load 3
	☐	(1.5) x Load 1
-	☐	111 : 0.9DL-1.5EQX
	☐	(0.9) x Load 3
	☐	(-1.5) x Load 1
-	☐	112 : 0.9DL+1.5EQZ
	☐	(0.9) x Load 3
	☐	(1.5) x Load 2
-	☐	113 : 0.9DL-1.5EQZ
	☐	(0.9) x Load 3
	☐	(-1.5) x Load 2

-	☐	114 : 1.2(DL+LL+WLX)
	☐	(1.2) x Load 3
	☐	(1.2) x Load 4
	☐	(1.2) x Load 5
-	☐	115 : 1.2(DL+LL-WLX)
	☐	(1.2) x Load 3
	☐	(1.2) x Load 4
	☐	(1.2) x Load 6
-	☐	116 : 1.2(DL+LL+WLZ)
	☐	(1.2) x Load 3
	☐	(1.2) x Load 4
	☐	(1.2) x Load 7

-	☐	117 : 1.2(DL+LL-WLZ)
	☐	(1.2) x Load 3
	☐	(1.2) x Load 4
	☐	(1.2) x Load 8
-	☐	118 : 1.5(DL+WLX)
	☐	(1.5) x Load 3
	☐	(1.5) x Load 5
-	☐	119 : 1.5(DL-WLX)
	☐	(1.5) x Load 3
	☐	(1.5) x Load 6
-	☐	120 : 1.5(DL+WLZ)
	☐	(1.5) x Load 3
	☐	(1.5) x Load 7

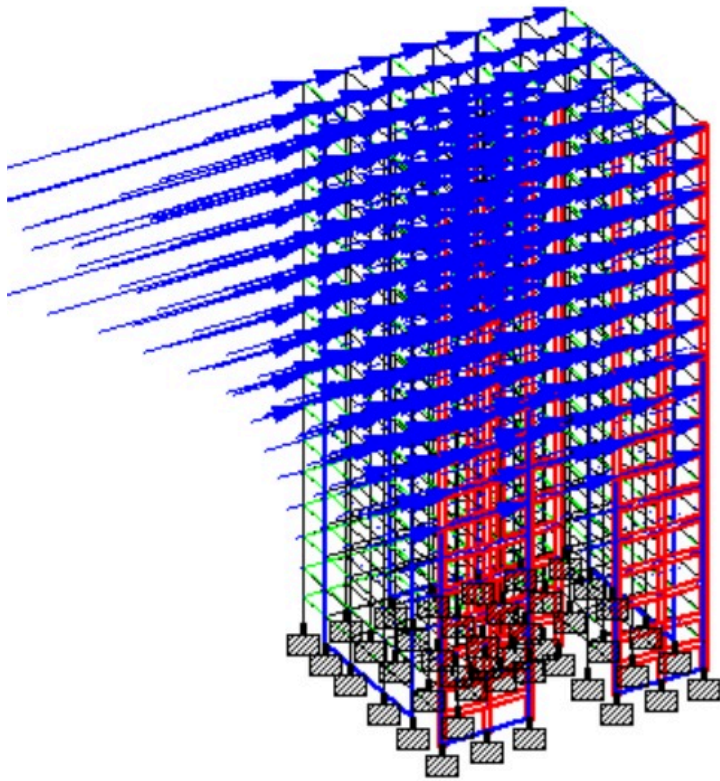
-	☐	121 : 1.5(DL-WLZ)
	☐	(1.5) x Load 3
	☐	(1.5) x Load 8
-	☐	122 : 0.9DL+1.5WLX
	☐	(0.9) x Load 3
	☐	(1.5) x Load 5
-	☐	123 : 0.9DL-1.5WLX
	☐	(0.9) x Load 3
	☐	(1.5) x Load 6
-	☐	124 : 0.9DL+1.5WLZ
	☐	(0.9) x Load 3
	☐	(1.5) x Load 7
-	☐	125 : 0.9DL-1.5WLZ
	☐	(0.9) x Load 3
	☐	(1.5) x Load 8

Support 1 Reaction forces for All Load Cases:

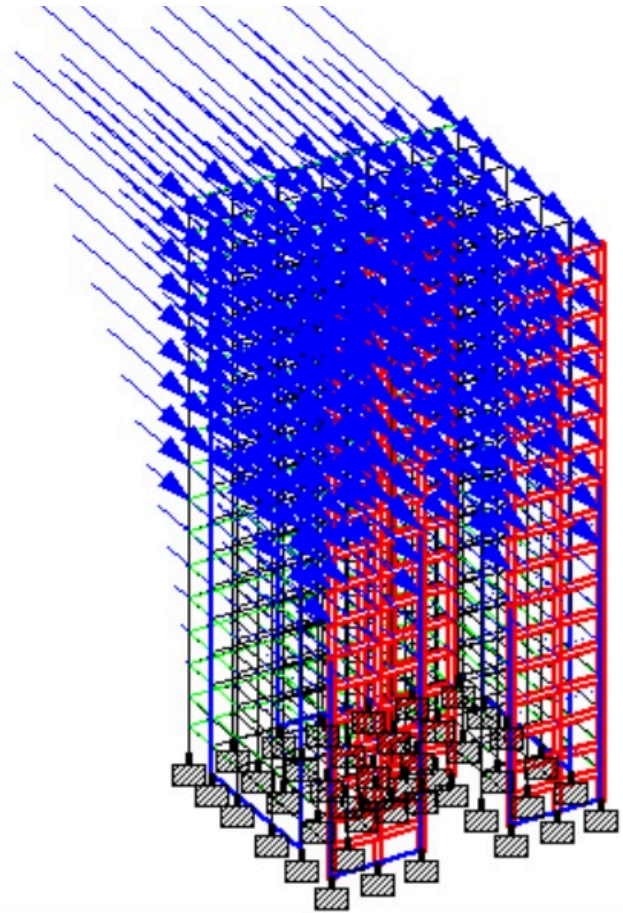
		Horizontal	Vertical	Horizontal	Moment		
Node	L/C	Fx kN	Fy kN	Fz kN	Mx kip-in	My kip-in	Mz kip-in
1	1 EQ X	-48.184	-601.515	-3.736	-72.583	-6.727	1208.571
	2 EQ Z	0.088	-738.696	-56.600	-1111.321	-0.071	-1.975
	3 DL	5.901	1370.766	1.964	-21.698	-4.417	-45.836
	4 LL	1.219	190.838	0.390	-4.917	-0.857	-9.583
	5 WL +X	-28.285	-270.184	-5.871	-112.036	-12.296	624.560
	6 WL -X	24.671	247.465	2.501	48.736	4.565	-609.067
	7 WL +Z	-0.055	-346.014	-38.245	-683.938	1.163	0.876
	8 WL -Z	-0.038	335.415	33.351	647.156	-5.407	1.049
	101 1.5(DL+L	10.680	2342.405	3.530	-39.922	-7.911	-83.128
	102 1.2(DL+L	-49.276	1152.106	-1.659	-119.037	-14.401	1383.782
	103 1.2(DL+L	66.365	2595.742	7.307	55.162	1.744	-1516.788
	104 1.2(DL+L	8.650	987.489	-65.095	-1365.523	-6.414	-68.873
	105 1.2(DL+L	8.439	2760.359	70.744	1301.648	-6.243	-64.132
	106 1.5(DL+E	-63.424	1153.876	-2.658	-141.421	-16.716	1744.102
	107 1.5(DL-E	81.127	2958.420	8.550	76.328	3.466	-1881.611
	108 1.5(DL+E	8.983	948.104	-81.954	-1699.528	-6.732	-71.717
	109 1.5(DL-E	8.720	3164.192	87.846	1634.435	-6.518	-65.792
	110 0.9DL+1.	-66.965	331.417	-3.836	-128.403	-14.066	1771.604
	111 0.9DL-1.	77.586	2135.961	7.371	89.347	6.116	-1854.109
	112 0.9DL+1.	5.442	125.645	-83.132	-1686.509	-4.082	-44.215
	113 0.9DL-1.	5.179	2341.733	86.667	1647.454	-3.868	-38.290
	114 1.2(DL+L	-25.398	1549.704	-4.221	-166.381	-21.083	682.969
	115 1.2(DL+L	38.149	2170.882	5.826	26.546	-0.850	-797.383
	116 1.2(DL+L	8.479	1458.708	-43.070	-852.663	-4.933	-65.451
	117 1.2(DL+L	8.498	2276.422	42.845	744.650	-12.817	-65.244
	118 1.5(DL+	-33.577	1650.873	-5.860	-200.601	-25.069	868.086
	119 1.5(DL-	45.857	2427.346	6.698	40.558	0.223	-982.355
	120 1.5(DL+	8.769	1537.128	-54.422	-1058.453	-4.881	-67.440
	121 1.5(DL-	8.794	2559.271	52.972	938.187	-14.736	-67.182
	122 0.9DL+1.	-37.117	828.413	-7.039	-187.582	-22.419	895.587
	123 0.9DL-1.	42.317	1604.887	5.519	53.576	2.873	-954.853
	124 0.9DL+1.	5.229	714.668	-55.600	-1045.434	-2.231	-39.938
	125 0.9DL-1.	5.253	1736.812	51.793	951.206	-12.086	-39.680

Analysis of structural system under load combinations:

Earthquake Load:

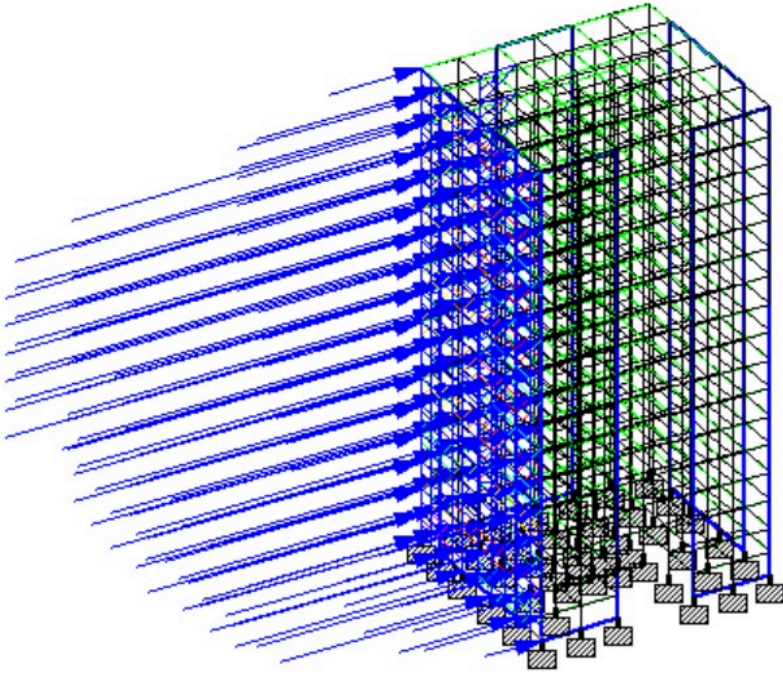


X-Direction

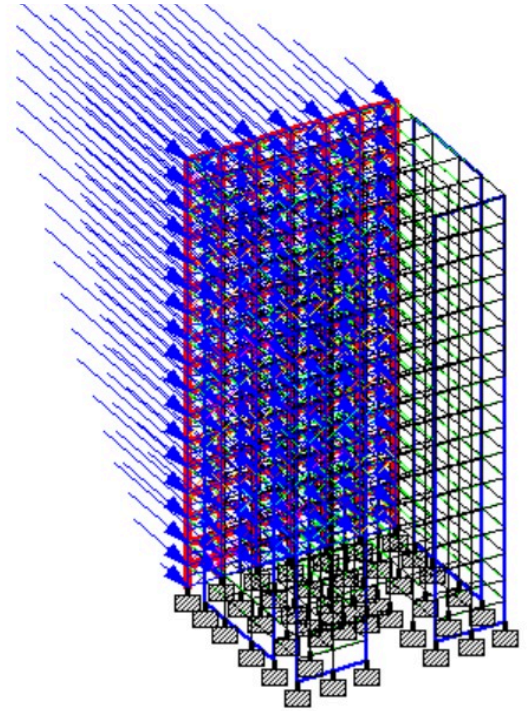


Z-Direction

Wind Load:



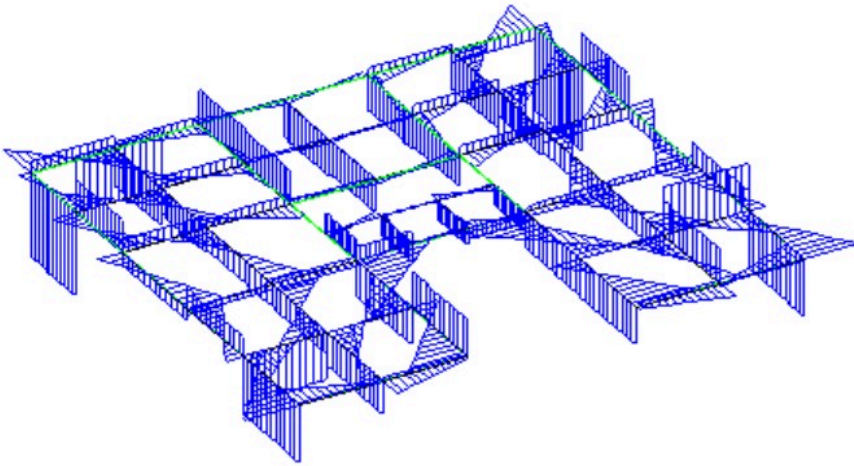
X-Direction



Z-Direction

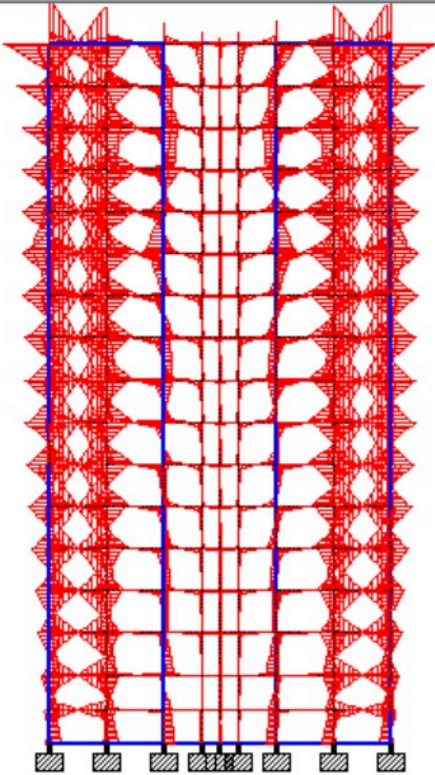
Shear and Bending moments diagram:

Shear Force Diagram:



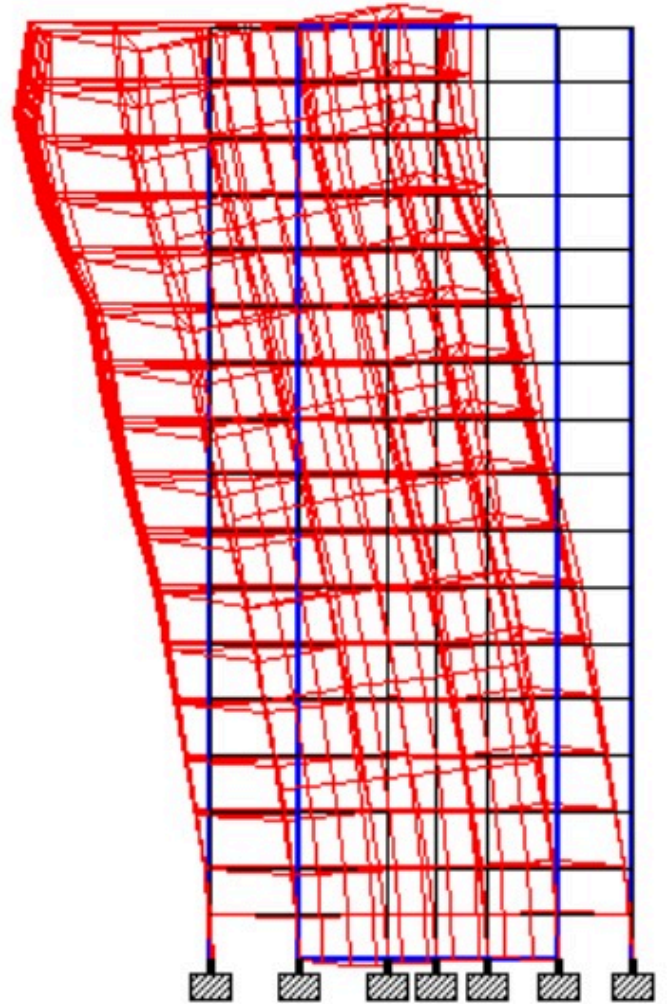
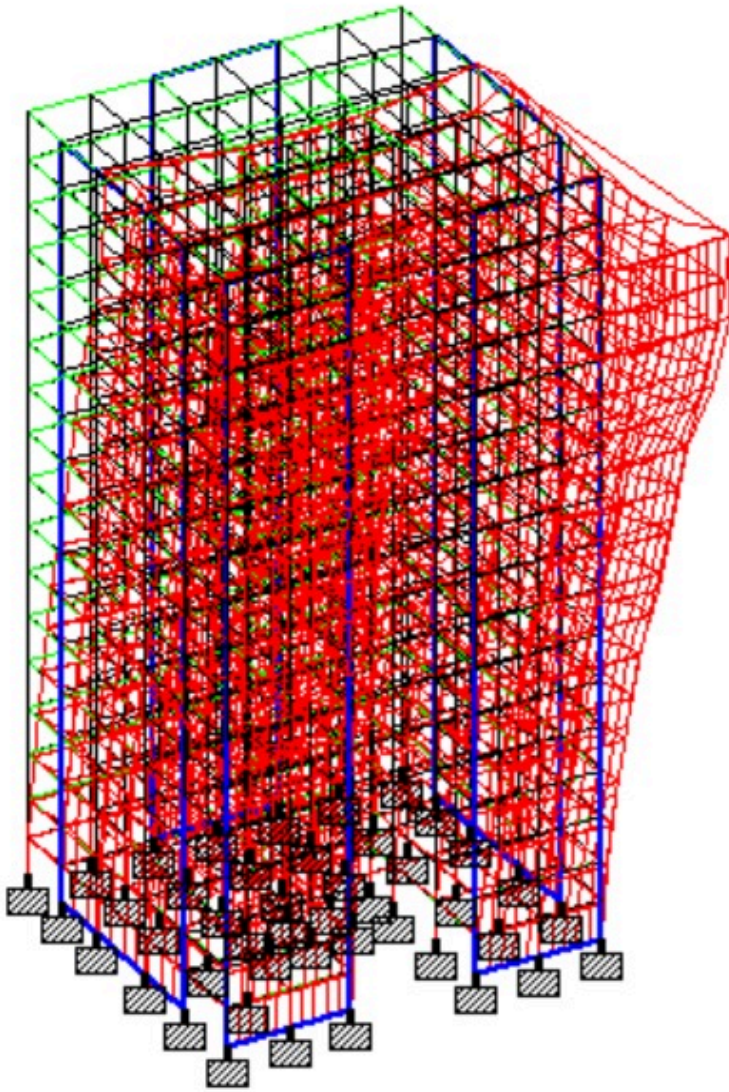
Shear Force Diagram per floor

Bending Moment Diagram:



Building Bending Moment Diagram

Building Deflections:



Minimum and Maximum Deflections:

			Horizontal	Vertical	Horizontal	Resultant	Rotational		
	Node	L/C	X mm	Y mm	Z mm	mm	rX rad	rY rad	rZ rad
Max X	788	106 1.5(DL+E	75.765	-5.991	1.668	76.020	-0.000	-0.001	-0.002
Min X	810	111 0.9DL-1.	-68.424	-5.009	-0.768	68.611	-0.000	0.000	0.001
Max Y	819	1 EQ X	41.147	3.989	-0.758	41.346	-0.000	-0.000	-0.001
Min Y	798	106 1.5(DL+E	75.708	-24.201	1.506	79.496	0.000	-0.000	-0.002
Max Z	826	108 1.5(DL+E	1.792	-9.710	46.595	47.629	0.001	0.000	-0.000
Min Z	806	113 0.9DL-1.	1.576	-3.895	-44.987	45.183	-0.000	-0.000	-0.000
Max rX	606	108 1.5(DL+E	1.758	-8.561	31.869	33.046	0.002	-0.000	-0.000
Min rX	617	109 1.5(DL-E	14.746	-10.704	-33.853	38.446	-0.002	0.000	-0.000
Max rY	788	109 1.5(DL-E	16.674	-1.337	-36.086	39.774	-0.000	0.002	-0.001
Min rY	811	109 1.5(DL-E	-7.507	-0.975	-38.260	39.001	-0.000	-0.002	0.001
Max rZ	632	111 0.9DL-1.	-44.186	0.612	-0.745	44.196	-0.000	0.000	0.002
Min rZ	609	106 1.5(DL+E	49.891	-2.489	0.894	49.961	-0.000	-0.000	-0.002
Max Rs	798	106 1.5(DL+E	75.708	-24.201	1.506	79.496	0.000	-0.000	-0.002

Minimum and Maximum Forces & Moments:

Design Sessional_final - Beam End Forces:									
All Summary Envelope									
	Beam	L/C	Node	Fx kN	Fy kN	Fz kN	Mx kN-m	My kN-m	Mz kN-m
Max Fx	23	106 1.5(DL+E	5	7021.851	37.875	6.751	0.525	-11.712	107.939
Min Fx	151	1 EQ X	44	-2014.225	38.462	4.265	0.334	-3.877	56.382
Max Fy	1765	101 1.5(DL+L	748	-13.637	208.929	-2.241	-0.101	4.724	235.738
Min Fy	186	101 1.5(DL+L	119	10.423	-204.534	0.186	0.171	0.210	225.096
Max Fz	1405	108 1.5(DL+E	605	935.119	-54.721	109.724	-2.150	-133.177	-81.820
Min Fz	1445	109 1.5(DL-E	619	867.485	81.530	-120.813	-1.591	170.603	122.728
Max Mx	62	107 1.5(DL-E	69	-23.112	79.722	-1.056	13.297	2.034	128.953
Min Mx	1411	106 1.5(DL+E	608	598.826	-21.664	14.029	-12.041	-25.564	-27.046
Max My	1405	108 1.5(DL+E	560	972.229	-54.721	109.724	-2.150	195.994	82.343
Min My	1445	109 1.5(DL-E	574	904.594	81.530	-120.813	-1.591	-191.835	-121.861
Max Mz	200	101 1.5(DL+L	121	82.034	182.483	0.057	-0.072	-0.209	296.604
Min Mz	1417	106 1.5(DL+E	566	1737.751	122.390	-60.126	0.422	-95.334	-215.088

Load Calculations:

Dead Load:

$$\text{Total Beams Vol} = 0.5 \times 0.35 \times (4.637 \times 12 + 4.05 \times 30 + 3.823 \times 14 + 5.423 \times 4 + 2.583 \times 4 + 2.7 \times 6) \times 17$$

$$= 829.69775 \text{ m}^3$$

$$\text{Total Column Volume} = 0.7 \times 0.5 \times 2.5 \times 45 \times 2 + 0.7 \times 0.5 \times 3 \times 45 \times 15$$

$$= 787.5 \text{ m}^3$$

$$\text{Self Weight} = (787.5 + 829.69775) \times 23.5616 \text{ KN}$$

$$= 38103.7665 \text{ KN}$$

$$\text{Total Floor Load} = \text{Floor Area} \times \text{No of floors} \times \text{floor load/floor}$$

$$= 467.8155 \times 16 \times 6.75 + 467.8155 \times 3.75 \text{ KN}$$

$$= 52278.38213 \text{ KN}$$

$$\text{Total Dead Load of Building} = 38103.7665 + 52278.38213 \text{ KN}$$

$$= 90382.14863 \text{ KN}$$

Verification with software:

$$\text{Total Reaction forces} = R1 + R2 + \dots + R45$$

$$= 93917.934 \text{ KN}$$

$$\text{Error\%} = ((93917.934 - 90382.14863) / 90382.14863) \times 100$$

$$= 3.912\%$$

Live Load:

$$\text{Total Live Load} = \text{Floor Area} \times \text{No of floors} \times \text{floor load/floor}$$

$$= 467.8155 \times 16 \times 2 + 467.8155 \times 1.5 \text{ KN}$$

$$= 15671.81925 \text{ KN}$$

Verification with software:

$$\begin{aligned}\text{Total LL Reaction forces} &= R1 + R2 + \dots + R45 \\ &= 16744.424 \text{ KN}\end{aligned}$$

$$\begin{aligned}\text{Error\%} &= ((16744.424 - 15671.81925) / 15671.81925) * 100 \\ &= 6.844\%\end{aligned}$$

Wind Load:

For Chandigarh, Wind = 47 m/s

$$\begin{aligned}\text{Design Factors: } K1 &= 1 && (\text{since 50 years design life, } V=47\text{m/s}) \\ K3 &= 1 && (\text{since slope} < 3\text{degrees considered}) \\ K4 &= 1 && (\text{since not a cyclonic region}) \\ K2 &= 1.5 && (\text{Terrain Category 2, Class B structure, } H_{\text{building}} = 50 \text{ m})\end{aligned}$$

$$\begin{aligned}V_z &= \text{Wind} * K1 * K2 * K3 * K4 \\ &= 1.15 * 47 \text{ m/s}\end{aligned}$$

$$P_z = 0.6 * V_z^2$$

$$\begin{aligned}F_{\text{wind}} &= C_p * \text{Area} * P_z \\ &= 0.8 * 0.6 * 47.5 * 22.343 * 1.15^2 * 47^2 \\ &= 1488.222 \text{ KN}\end{aligned}$$

Verification with software:

$$\begin{aligned}\text{Total WL Reaction forces} &= R1 + R2 + \dots + R45 \\ &= 1632.012 \text{ KN}\end{aligned}$$

$$\text{Error\%} = ((1632.012 - 1488.222) / 1488.222) * 100 = 9.661\%$$

Earthquake Load:

Static Analysis Method: $V_b = A_h * W$

$$A_h = ((z/2) * (S_a/g)) / (R/I)$$

$$(z = 0.24, I = 1.2, R = 5)$$

$$T_a = 0.09 * H / d^{1/2}$$

$$\text{soil bearing capacity} = 12 \text{ ton/m}^2 \Rightarrow \text{soft soil}$$

$$\Rightarrow T_{ax} = 0.09 * 47.5 / 24.3^{1/2} = 0.8672$$

$$T_{az} = 0.09 * 47.5 / 22.34^{1/2} = 0.9044$$

For $0.67 < T < 4$ secs & for soft soil, $S_a/g = 1.67/T_a$

$$\Rightarrow (S_a/g)_x = 1.67 / 0.8672 = 1.925 \Rightarrow A_{hx} = 0.05544$$

$$(S_a/g)_z = 1.67 / 0.9044 = 1.8464 \Rightarrow A_{hz} = 0.05317$$

$$W = D_I + x\% \text{ of } LL \quad (x=25\% \text{ for } LL < 3 \text{ KN/m}^2)$$

$$= 90382.14863 + 0.25 * 15671.81925 \text{ KN}$$

$$= 94300.10344 \text{ KN}$$

$$\text{Earthquake design shear in x-direction} = A_{hx} * W = 0.05544 * 94300.10344$$

$$= 5227.997735 \text{ KN}$$

$$\text{Earthquake design shear in z-direction} = A_{hz} * W = 0.05317 * 94300.10344$$

$$= 5013.9365 \text{ KN}$$

Verification with Software

$$\text{In x-direction} = R_1 + R_2 + \dots + R_{45} = 2719.564 \text{ KN}$$

$$\text{In z-direction} = R_1 + R_2 + \dots + R_{45} = 2719.564 \text{ KN}$$

Design Calculations

Beam Design:

1. Design Data Extracted from [STAAD.Pro](#)

- Beam Number: 77
- Width (b): 350 mm
- Total Depth (D): 400 mm
- Clear Span (L): 4636 mm
- Concrete Grade (fck): 30 MPa (M30)
- Steel Grade (fy): 500 MPa (Fe500)
- Max Bending Moment (Mu): 75.143 kN-m (from Critical LC 108)
- Max Shear Force (Vu): 55.736 kN (from Critical LC 109)

Design Sessional_final - Beam

Geometry Property Loading Shear Bending Deflection Concrete Design

Beam no. = 77. Section: Rect 15.75x13.78

0.400m

Length = 4.636m

0.350m

Section diagrams are approximate representation and not to the scale.

Physical Properties

Ax(m2)	0.140	Ix(m4)	0.003
Ay(m2)	0.117	Iy(m4)	0.001
Az(m2)	0.117	Iz(m4)	0.002
D(m)	0.400	W(m)	0.350

Assign/Change Property

Material Properties

Elasticity(MPa)	21718.594	Density(kg/m3)	2402.630
Poisson	0.17	Alpha	1e-05

CONCRETE

Assign Material

Design Sessional_final - Beam

Geometry Property Loading Shear Bending Deflection Concrete Design

Beam no. = 77 Design code : IS-456

5#12 @ 354.00 0.00 To 3090.67 5#12 @ 354.00 3090.67 To 4636.00

15 # 8 c/c 150.00 15 # 8 c/c 150.00

3#12 @ 46.00 0.00 To 4636.00

at 0.000 at 2318.000 at 4635.999

Design Load

Mz Kn Met	Dist. Met	Load
26.55	3.1	109
-74.49	0	109
-75.14	4.6	108

Design Parameter

Fy(Mpa)	500
Fc(Mpa)	30
Depth(m)	0.400
Width(m)	0.350
Length(m)	4.636

Print Close

Geometry Property Loading Shear Bending Deflection Concrete Design

Beam No = 77

75.14

72-5.57 3.05 73

Section Forces

Dist. m	Fy kN	Mz kN-m
2.704	-25.329	-11.041
3.091	-36.323	1.280
3.477	-41.803	16.528
3.863	-48.904	34.401
4.250	-53.410	54.084

Dist. m Fy kN Mz kN-m

0.000 20.826 -5.571

Selection Type

Load Case : 108:1.5(DL+EQZ)

Bending - Z Bending - Y

2. Design for Flexure (Main Reinforcement)

2.1 Effective Depth Calculation

Assume:

- Clear cover = 30 mm
- Main bar diameter = 12 mm

$$d' = 30 + 12/2 = 36 \text{ mm}$$

$$d = 400 - 36 = 364 \text{ mm}$$

2.2 Check Limiting Moment Capacity

For Fe500 steel: $x_{u,max}/d = 0.46$

$$M_{u,lim} = 0.133 \times f_{ck} \times b \times d^2 = 0.133 \times 30 \times 350 \times (364)^2 = 184.92 \times 10^6 \text{ N-mm} = 184.92 \text{ kN-m}$$

Since: $M_u(75.143 \text{ kN-m}) < M_{u,lim}(184.92 \text{ kN-m})$

The section is under-reinforced, safe, and ductile.

2.3 Required Area of Steel (A_{st})

$$A_{st} = \left(0.5f_{ck}/f_y \right) \left[1 - \sqrt{1 - 4.6M_u/f_{ck}bd^2} \right] bd$$

$$A_{st} = 508.6 \text{ mm}^2$$

2.4 Check Codal Limits

$$A_{st,min} = 0.85bd/f_y = 216.58 \text{ mm}^2$$

$$A_{st,max} = 0.04bD = 5600 \text{ mm}^2$$

$$216.58 < 508.6 < 5600 \Rightarrow \text{OK}$$

2.5 Provide Reinforcement

- Area of 12 mm bar = 113 mm²

$$\text{No. of bars} = 508.6 / 113 = 4.5$$

Provide 5 bars of 12 mm dia

$A_{st,prov} = 565 \text{ mm}^2$

(Matches STAAD output: 5#12)

3. Design for Shear (Stirrups)

3.1 Nominal Shear Stress

$$\tau_v = V_u/b \times d = 55736/350 \times 364 = 0.437 \text{ MPa}$$

3.2 Design Shear Strength of Concrete

$$p_t = 100 \times 565/350 \times 364 = 0.444\%$$

From IS 456 Table 19 (M30):

• At 0.25% $\rightarrow$ 0.37 MPa • At 0.50% $\rightarrow$ 0.50 MPa

Interpolated: $\tau_c = 0.47 \text{ MPa}$

Since: $\tau_v(0.437) < \tau_c(0.47)$

Only minimum shear reinforcement is required.

3.3 Minimum Stirrup Design

Assume 2-legged 8 mm stirrups: $A_{sv} = 2 \times \pi/4 \times 8^2 = 100.53 \text{ mm}^2$

$$S_v = 0.87 f_y A_{sv} / 0.4 b$$

$$S_v = 0.87 \times 500 \times 100.53 / 0.4 \times 350 = 312.3 \text{ mm}$$

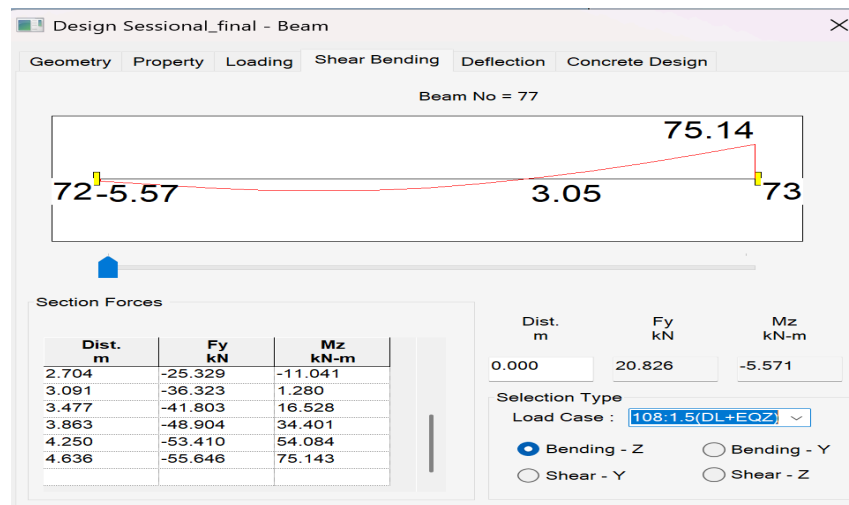
3.4 Maximum Spacing Check

Maximum spacing = least of: • $0.75d = 273 \text{ mm}$ (governs) • 300 mm • 312.3 mm

Provide: 2-legged 8 mm stirrups @ 250 mm c/c

Note STAAD provided 150 mm spacing due to ductile detailing requirements as per IS 13920 (earthquake design).

For IS 456 gravity design, 250 mm c/c is safe.



4. Deflection Check

Deflection is a "Serviceability Limit State" check. According to IS 456, we check deflection for normal, everyday loads without the 1.5 safety factor so for Load combination 126 $1.0(DD+LL)$.

4.1 Maximum Permissible Deflection

As per IS 456:2000,

the final deflection should not exceed:

Permissible Limit = $\min (L/250, 20 \text{ mm}) = \min (4636/250, 20 \text{ mm}) = \min (18.54 \text{ mm}, 20 \text{ mm})$

Max Allowable Deflection = 18.54 mm

4.2 Actual Deflection from Analysis

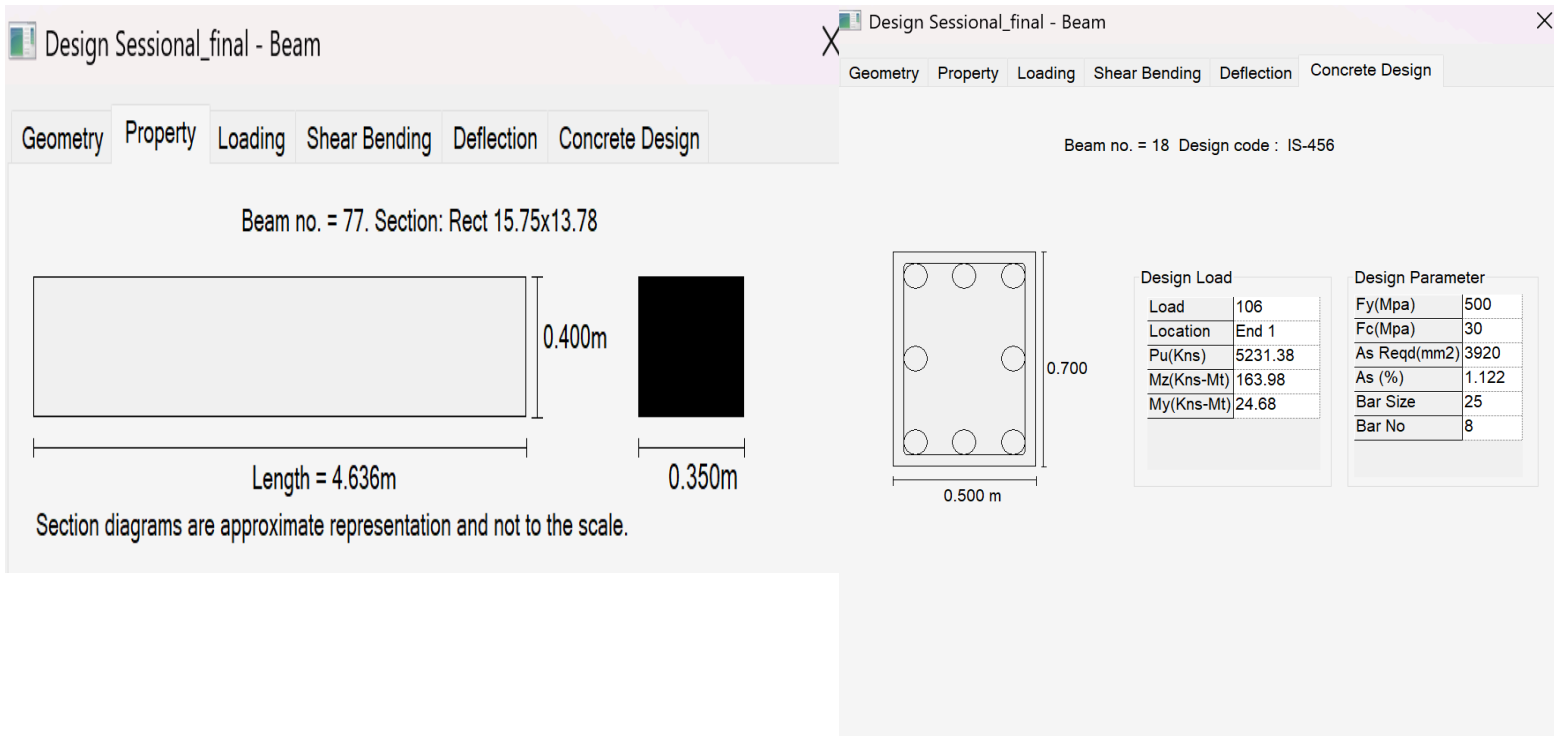
Maximum displacement from STAAD (Service Load Combination 126: DL + LL): = 1.0 mm

4.3 Conclusion

Actual Deflection(1.0 mm) < Permissible Deflection(18.54 mm)

Hence, the beam deflection is well within the permissible limits.

Design Of Column:-



MANUAL CALCULATIONS AND RESULTS (As per IS 456:2000 & SP 16)

1. Design Data (Beam No. 18)

$P_u = 5231.38 \text{ kN}$
 $M_z = 163.98 \text{ kNm}$ (Major axis)
 $M_y = 24.68 \text{ kNm}$ (Minor axis)
 $L = 2.5 \text{ m} = 2500 \text{ mm}$
 $b = 500 \text{ mm}, D = 700 \text{ mm}$
 $f_{ck} = 30 \text{ MPa}, f_y = 500 \text{ MPa}$

2. Check for Minimum Eccentricity Moments (IS 456 Cl. 25.4)

$$e_{min,z} = L/500 + D/30 = 2500/500 + 700/30 = 5 + 23.33 = 28.33 \text{ mm} (> 20 \text{ mm})$$

$$e_{min,y} = L/500 + b/30 = 2500/500 + 500/30 = 5 + 16.67 = 21.67 \text{ mm } (> 20 \text{ mm})$$

Moments due to Minimum Eccentricity

$$M_{z,min} = P_u \times e_{min,z} = 5231.38 \times 0.02833 = 148.2 \text{ kNm}$$

$$M_{y,min} = P_u \times e_{min,y} = 5231.38 \times 0.02167 = 113.36 \text{ kNm}$$

Design Moments

$$M_z = \max(163.98, 148.2) = 163.98 \text{ kNm}$$

$$M_y = \max(24.68, 113.36) = 113.36 \text{ kNm}$$

3. Calculation of Required A_{sc} (SP 16 Method)

$$\text{Equivalent Moment(Approximate method)} M_{u,eq} = M_z + (D/b)$$

$$M_y = 163.98 + (700/500) \times 113.36 = 163.98 + 158.70 = 322.68 \text{ kNm}$$

Non-Dimensional Parameters

$$P_u/f_{ck} \cdot b \cdot D = 5231.38 \times 10^3/30 \times 500 \times 700 = 0.498$$

$$M_{u,eq}/(f_{ck} \cdot b \cdot D^2) = 322.68 \times 10^6/30 \times 500 \times 700^2 = 0.044$$

From SP 16 Chart 44 (Fe 500)

$$p/f_{ck} \approx 0.037$$

$$p = 0.037 \times 30 = 1.11\%$$

$$\text{Required Steel Area } A_{sc,reqd} = (1.11/100) \times 500 \times 700 = 3885 \text{ mm}^2$$

Provided Reinforcement

Provide 8 bars of 25 mm diameter:

$$A_{sc,prov} = 8 \times 490.87 = 3927 \text{ mm}^2$$

$$p = 1.122\%$$

4. Biaxial Interaction Check (IS 456 Cl. 39.6)

$$p/f_{ck} = 1.122/30 = 0.0374$$

From SP 16:

$$X \approx 0.058$$

Moment Capacities

$$M_{uz1} = 0.058 \times 30 \times 500 \times 700^2 \times 10^{-6} = 426.3 \text{ kNm}$$

$$M_{uy1} = 0.058 \times 30 \times 700 \times 500^2 \times 10^{-6} = 304.5 \text{ kNm}$$

Axial Capacity

$$P_{uz} = 0.45f_{ck}A_c + 0.75f_yA_{sc} =$$

$$0.45 \times 30 \times (350000 - 3927) + 0.75 \times 500 \times 3927 \\ = 6144.6 \text{ kN}$$

$$P_u/P_{uz} = 5231.38/6144.6 = 0.851$$

$$\alpha_n = 2.0$$

Interaction Equation

$$(M_z/M_{uz1})^2 + (M_y/M_{uy1})^2$$

$$= (163.98/426.3)^2 + (113.36/304.5)^2$$

$$= 0.148 + 0.138 = 0.286 \leq 1.$$

Conclusion

Design is SAFE.

Final Comparison

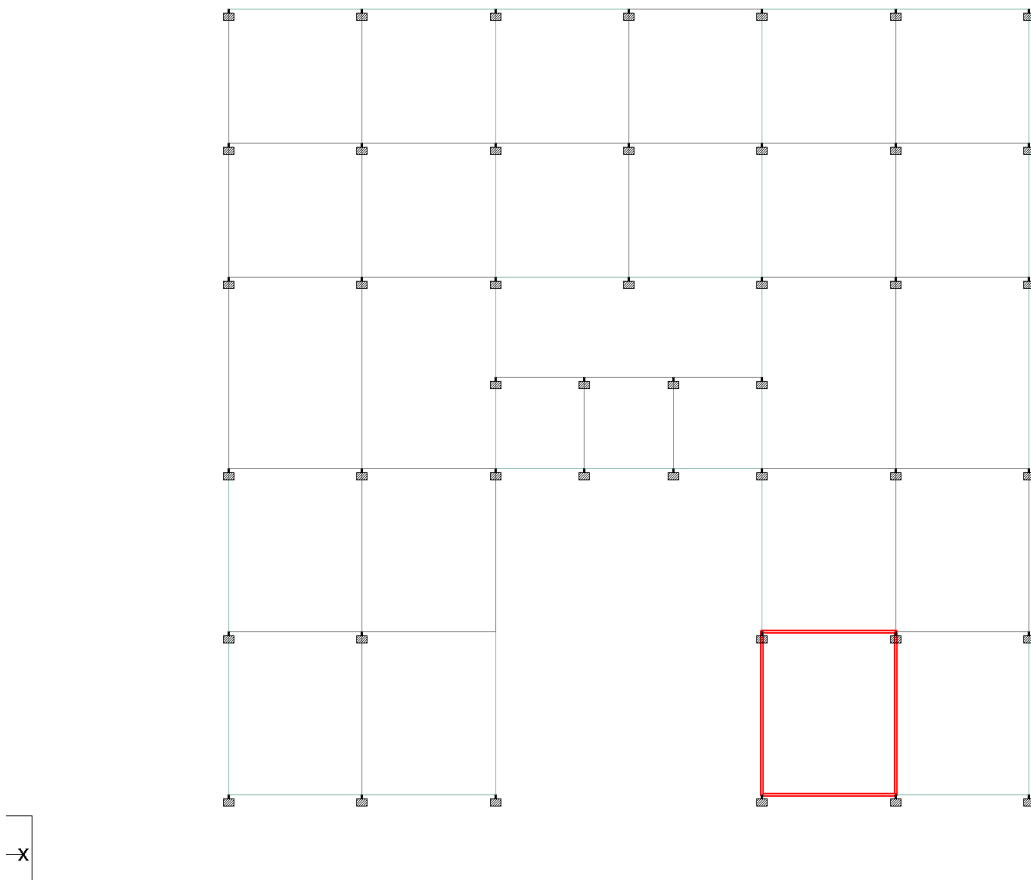
$$A_{sc} \text{ (STAAD)} = 3920 \text{ mm}^2$$

$$A_{sc} \text{ (Manual)} = 3885 \text{ mm}^2$$

$$A_{sc} \text{ (Provided)} = 3927 \text{ mm}^2$$

Design of Slab

MANUAL CALCULATIONS AND RESULTS: TWO-WAY SLAB



Design Data

Slab adjacent to Beam $L_x = 4.05$ m(Short span)

$L_y = 4.637$ m(Long span)

$L_y/L_x = 4.637/4.05 = 1.145$

As per IS 456:2000, Annex D, Clause D-1.1:

Since: $L_y/L_x < 2.0$

Slab is designed as a Two-Way Slab

Materials:

- Concrete: M30
- Steel: Fe500

Step 1: Trial Depth Calculation As per IS 456:2000, Clause 23.2.1:

$$L/d = 26 \text{ (continuous slab)}$$

Assuming Modification Factor = 1.2

$$d_{req} = 4050/26 \times 0.8 = 194.7 \text{ mm}$$

Assume overall depth: $D = 220 \text{ mm}$

$$\text{Effective depth: } d = 220 - 20 - 8/2 = 196 \text{ mm}$$

Step 2: Load Calculation

$$\text{Self-weight: } 0.22 \times 25 = 5.50 \text{ kN/m}^2$$

$$\text{Floor finish: } 1.50 \text{ kN/m}^2$$

$$\text{Live load: } 2.00 \text{ kN/m}^2$$

$$TL = 9.00 \text{ kN/m}^2$$

$$\text{Factored load: } W = 1.5 \times 9 = 13.5 \text{ kN/m}^2$$

Step 3: Bending Moments (IS 456:2000, Annex D, Table 26)

Condition:

Two adjacent edges discontinuous

$$\alpha_x = 0.056, \alpha_y = 0.047$$

$$M_x = 0.056 \times 13.5 \times 4.052 = 12.40 \text{ kNm}$$

$$M_y = 0.047 \times 13.5 \times 4.052 = 10.40 \text{ kNm}$$

Step 4: Depth Check from Moment

$$M_{u,lim} = 0.133 \times f_{ck} \times b \times d^2$$

$$12.40 \times 10^6 = 0.133 \times 30 \times 1000 \times d^2$$

$$d_{req} = 55.7 \text{ mm} \text{ Since: } 55.7 < 196$$

Slab is safe in flexure.

Step 5: Shear Check

$$V_u = 0.5 \times 13.5 \times 4.05 = 27.33 \text{ kN}$$

$$\tau_v = 27.33 \times 10^3 / 1000 \times 196 = 0.139 \text{ N/mm}^2$$

From IS 456 Table 19: $\tau_c = 0.29 \text{ N/mm}^2$

Since: $0.139 < 0.29$ No shear reinforcement required.

STEP 6: CALCULATION OF AREA OF STEEL

(A_{st}) Steel Required for Flexure ($A_{st,req}$)

Critical bending moment (short span): $M_x = 12.40 \text{ kNm}$

Using IS 456:2000 (Annex G): $A_{st,req} = 0.5 f_{ck} / f_y [1 - \sqrt{1 - 4.6 M_u / f_{ck} b d^2}] b d$

Given Data

$$M_u = 12.40 \times 10^6 \text{ N-mm}$$

$$f_{ck} = 30 \text{ MPa}, f_y = 500 \text{ MPa}$$

$$b = 1000 \text{ mm}, d = 196 \text{ mm}$$

Calculation

$$\begin{aligned} A_{st,req} &= 0.5 \times 30 / 500 [1 - \sqrt{1 - (4.6 \times (12.40 \times 10^6) / 30 \times 1000 \times (196)^2)}] \times 1000 \times 196 \\ &= 0.03 [1 - \sqrt{1 - 57.04 \times 10^6 / 1,152,480,000}] \times 196000 = 0.03 [1 - \sqrt{1 - 0.0495}] \times 196000 \\ &= 0.03 [1 - 0.9749] \times 196000 = 0.03 \times 0.0251 \times 196000 \end{aligned}$$

$$A_{st,req} = 147.6 \text{ mm}^2 / \text{m width}$$

Since:

$$A_{st,req} < A_{st,min}$$

Minimum reinforcement governs the design

Final Conclusion • Provide:

$A_{st} = 264 \text{ mm}^2 / \text{m width}$

Since the maximum moment M_x does not exceed the minimum steel requirement, the smaller moment: $M_y = 10.40 \text{ kNm}$ also remains within limits.

STEP 7: REINFORCEMENT SPACING & DETAILING

1. Spacing Calculation

Governing steel area: $A_{st,req} = 264 \text{ mm}^2 / \text{m}$

Adopted bar diameter: $\phi = 8 \text{ mm}$ Area of one bar: $a_{st} = \pi/4 \times 8^2 = 50.26 \text{ mm}^2$

Required Spacing $s_{req} = 1000 \times a_{st}/A_{st,req} = 1000 \times 50.26/264 = 190.3 \text{ mm c/c}$

2. Maximum Spacing Check (IS 456:2000, Cl. 26.3.3 b.1)

Maximum allowable spacing = min of: $3d = 3 \times 196 = 588 \text{ mm}, 300 \text{ mm}$

Maximum spacing = **300 mm**

Adopt practical spacing: **175 mm c/c** Check: $175 < 300 \Rightarrow \text{OK}$

Actual Steel Provided:- $A_{st,prov} = 1000 \times 50.26/175 = 287 \text{ mm}^2 / \text{m}$

$287 > 264 \Rightarrow \text{OK}$

Provide 8 mm diameter bars @ 175 mm c/c both ways (bottom)

Design of whole structure with shear walls:

The thickness of the shear wall assumed is 200mm.

Maximum Drift Calculation

(After adding shear wall)

From Staad pro:-

Maximum deflection (δ) = 118.594 mm

Height of building (h) = 50 m

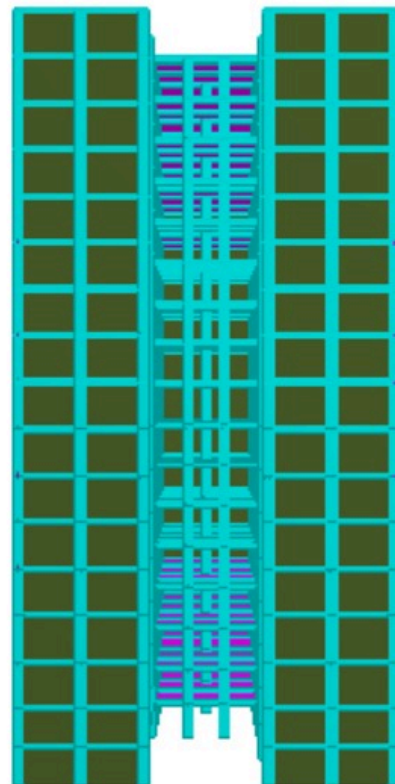
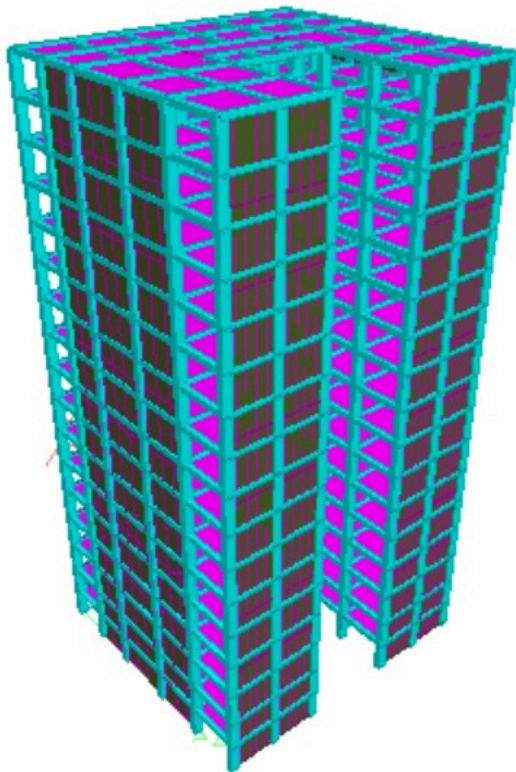
Maximum drift = δ/h

$$= (118.594/50) \times (10^{-3})$$

$$= 0.00237188$$

As per IS 1893 (Part 1) - 2016 Cl. 7.11.1.1, Maximum drift ≤ 0.004

Therefore, the building is safe against deflection.



Design and Detailing of Foundation:

Column Size = 0.7 m × 0.5 m

Safe Bearing Capacity (SBC) = 100 kN/m²

Grade of Concrete = M30

Grade of Steel = Fe500

Unit Weight of Concrete = 30 kN/m³

Factored Axial Load (P) = 1561.603 kN

DESIGN OF FOOTING SIZE

Required area of footing:

$$A = P / \text{SBC}$$

$$A = 1561.603 / 100$$

$$A = 15.61 \text{ m}^2$$

Assume square footing:

$$L = B = \sqrt{15.61} = 3.95 \text{ m}$$

Provide:

Footing size = 4.2 m × 4.2 m

Area provided = 17.64 m²

Since the provided area is greater than required area, the footing is safe.

SOIL PRESSURE CHECK

$$q = P / A$$

$$q = 1561.603 / 17.64$$

$$q = 88.55 \text{ kN/m}^2$$

Maximum pressure from analysis = 98.54 kN/m²

Since maximum pressure is less than SBC (100 kN/m²), the footing is safe.

DEPTH OF FOOTING

Overall depth (D) = 0.656 m

Effective depth (d) = 0.302 m

BENDING MOMENT CHECK

Ultimate moment (M_u) = 573.47 kNm

Limiting moment capacity ($M_{u,lim}$) = 1525.21 kNm

Since $M_u < M_{u,lim}$, the section is safe in bending.

SHEAR CHECK

One-Way Shear:

Shear stress = 275.68 kN/m²

Permissible shear = 277.55 kN/m²

Since actual shear stress is less than permissible, it is safe.

Two-Way (Punching) Shear:

Shear stress = 743.46 kN/m²

Permissible shear = 1369.30 kN/m²

Since actual shear stress is less than permissible, it is safe.

REINFORCEMENT DESIGN

Bottom Reinforcement

Along X-direction:

Provide 10 mm diameter bars at 75 mm c/c

Along Z-direction:

Provide 10 mm diameter bars at 65 mm c/c

Top Reinforcement

Provide 6 mm diameter bars at 60 mm c/c

DEVELOPMENT LENGTH CHECK

Required development length = 1.418 m

Since available length is greater than required, it is safe.

FINAL DETAILING

Footing Size = 4.2 m × 4.2 m

Thickness = 656 mm

Clear cover = 50 mm

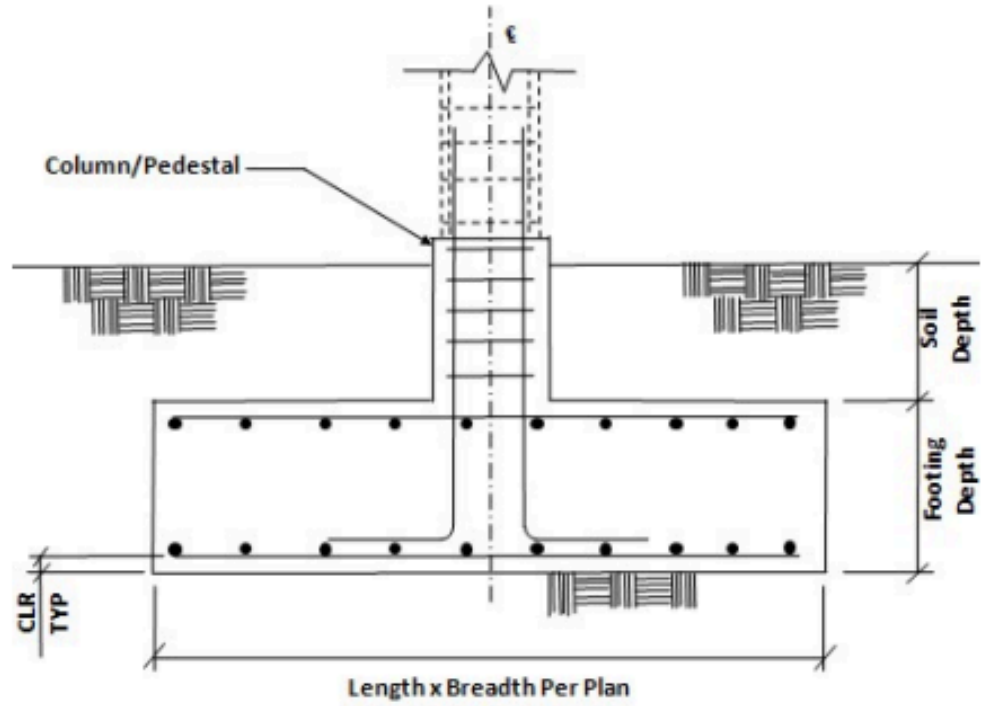
Reinforcement Details:

Bottom Steel:

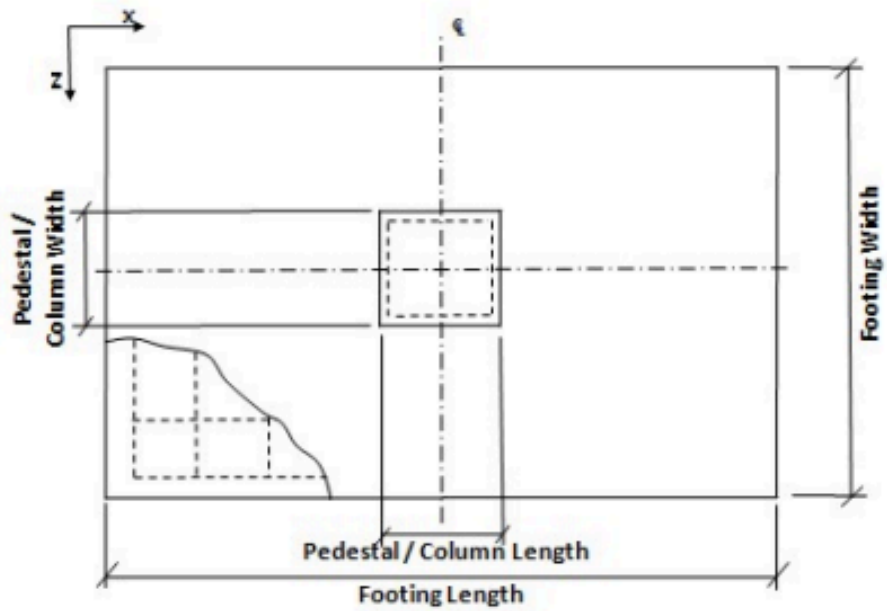
- X-direction → 10 mm @ 75 mm c/c
- Z-direction → 10 mm @ 65 mm c/c

Top Steel:

- 6 mm @ 60 mm c/c



ELEVATION



PLAN

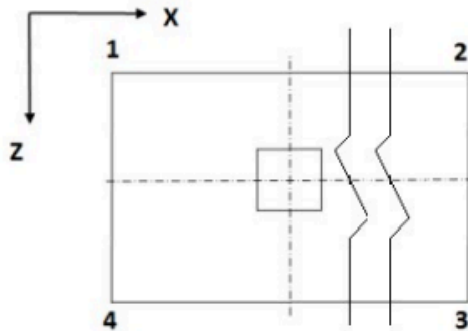
Final Footing Size

Length (L_2) = 4.200 m Governing Load Case : # 101

Width (W_2) = 4.200 m Governing Load Case : # 101

Depth (D_2) = 0.656 m Governing Load Case : # 101

Area (A_2) = 17.640 m²



Load Case	Pressure at corner 1 (q_1) (kN/m ²)	Pressure at corner 2 (q_2) (kN/m ²)	Pressure at corner 3 (q_3) (kN/m ²)	Pressure at corner 4 (q_4) (kN/m ²)	Area of footing in uplift (A_u) (m ²)
101	98.1738	96.8079	97.1787	98.5446	0.000
101	98.1738	96.8079	97.1787	98.5446	0.000
101	98.1738	96.8079	97.1787	98.5446	0.000
101	98.1738	96.8079	97.1787	98.5446	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

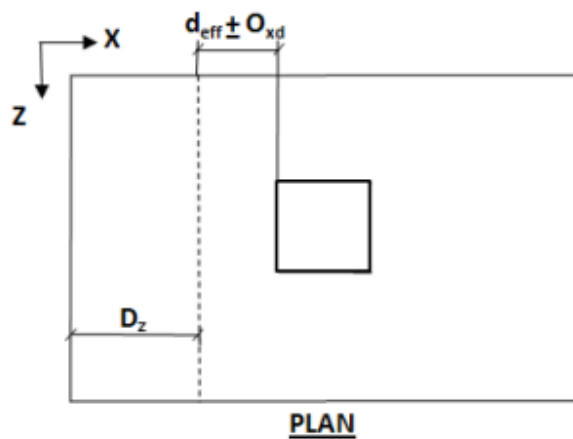
Check Trial Depth for one way shear (Along Z Axis)
(Shear Plane Parallel to Z Axis)

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26-03-26

Isolated Footing Design

Page 10 of 532



Critical Load Case = #201

$$D_z = 0.602 \text{ m}$$

$$\text{Shear Force}(S) = 643.847 \text{ kN}$$

$$\text{Shear Stress}(T_v) = 254.646063 \text{ kN/m}^2$$

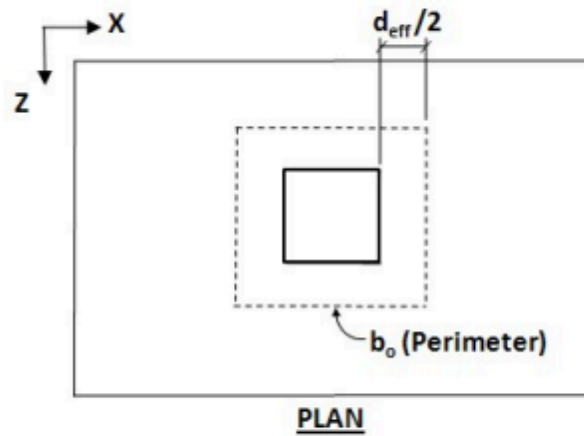
$$\text{Percentage Of Steel}(P_c) = 0.1480$$

As Per IS 456 2000 Clause 40 Table 19

$$\text{Shear Strength Of Concrete}(T_c) = 291.760 \text{ kN/m}^2$$

$T_v < T_c$ hence, safe

Check Trial Depth for two way shear



Critical Load Case = #201

Shear Force(S) = 2151.879 kN

Shear Stress(T_v) = 743.459 kN/m²

As Per IS 456 2000 Clause 31.6.3.1

$$K_s = \min[(0.5 + \beta), 1] = 1.000$$

$$\text{Shear Strength}(T_c) = 0.25 \times \sqrt{f_{ck}} = 1369.3064 \text{ kN/m}^2$$

$$K_s \times T_c = 1369.3064 \text{ kN/m}^2$$

$$T_v \leq K_s \times T_c \text{ hence, safe}$$

Top Reinforcement Design

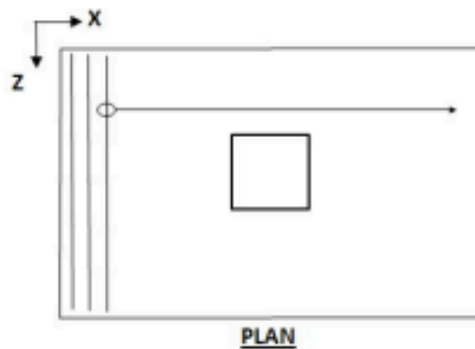
Along Z Axis

file:///C:/Staad.foundation%205.3/CalcXsl/footing.xml

26-03-26

Isolated Footing Design

Page 14 of 532



Minimum Area of Steel (A_{stmin}) = 1799.280 mm²

Calculated Area of Steel (A_{st}) = 1789.200 mm²

Provided Area of Steel ($A_{st,Provided}$) = 1799.280 mm²

$A_{stmin} \leq A_{st,Provided}$ Steel area is accepted

Governing Moment = 73.069 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 64.984 mm

$S_{min} \leq S \leq S_{max}$ and selected bar size < selected maximum bar size...

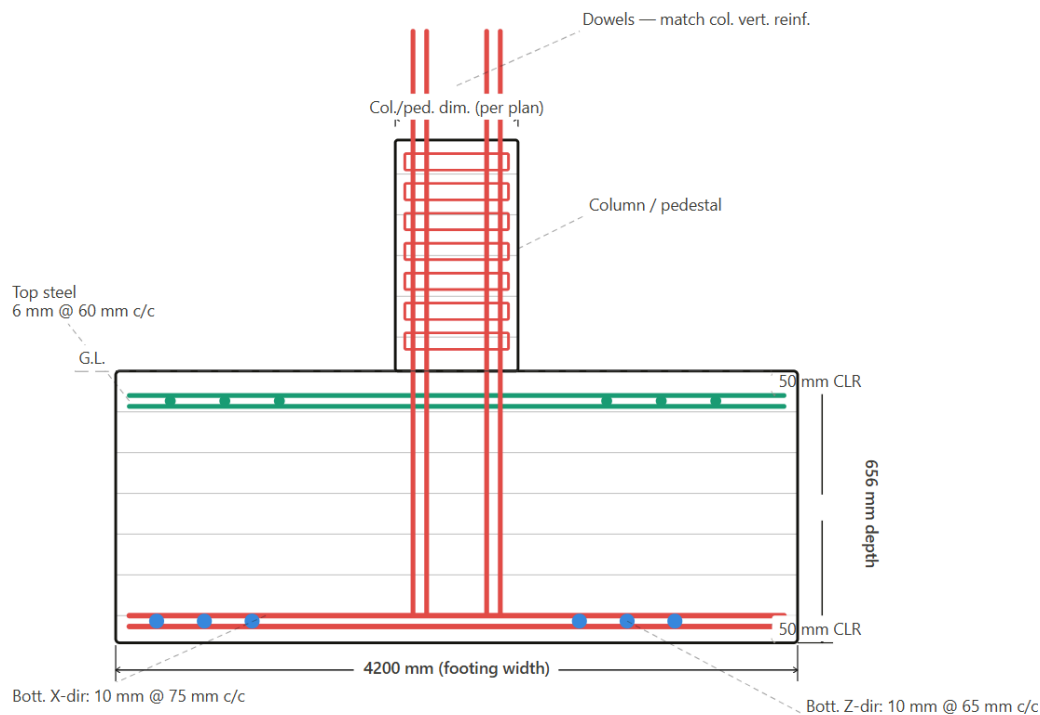
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

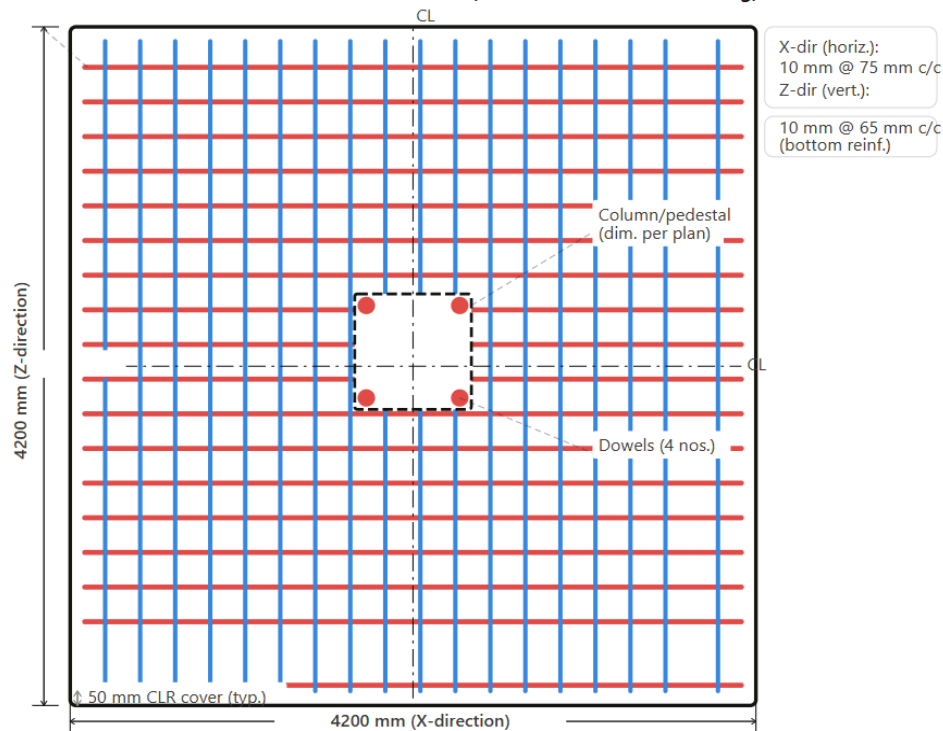
Ø6 @ 60 mm o.c.

Final reinforcement detailing:

Elevation — isolated footing (4200 mm × 4200 mm × 656 mm thick)



Plan view — bottom reinforcement (4200 mm × 4200 mm footing)



Beam Column Junction:

As per IS-456-2000 cl. 26.2.1,

$$\text{Development Length}(L_d) = (\phi * \sigma) / (4 * T_{bd})$$

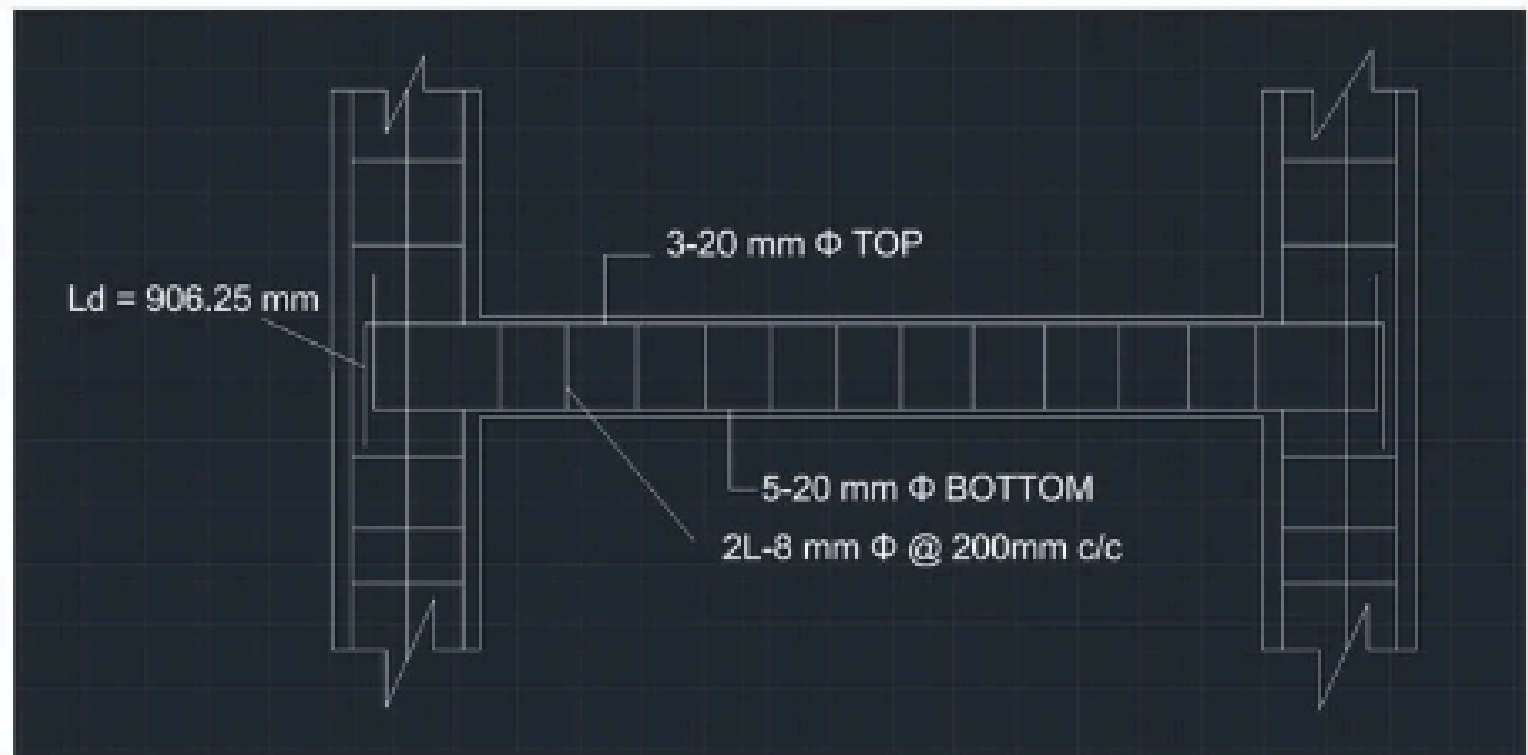
$$\sigma = 0.87f_y = 0.87 * 500 = 435 \text{ N/mm}^2$$

$$\phi = d = 20 \text{ mm}$$

For deformed bars, M30 concrete,

$$T_{bd} = 1.5 * 1.6 = 2.4 \text{ N/mm}^2$$

$$\text{So } L_d = (435 * 20) / (4 * 2.4) = 906.25 \text{ mm}$$



Plumbing Network:

Water Supply Design:

As per IS 1172:

- Water Demand = 135 LPCD (Domestic use) + 45 LPCD (Flushing requirements)
= 180 LPCD

- There are 5 apartments on each floor (three studio apartments, two 2-BHK apartments).

The building is a G+15 floors apartment.

We assume each apartment has a family of 4 members (excluding ground floor)

- Total Population = $15 * 5 * 4 = 300$

- Let's assume a factor of safety of 1.5 to account for emergency needs and varying water needs

- Total Water Demand = $1.5 * 300 * 180 = 81000$ litres/day

- Let's assume we supply water to the apartment in the morning and evening for 4 hours each. So total duration of water pumping to fill water tank = 8 hours

Rate of pumping = $(81000/8)$ Litres/Hour = 10125 Litres/Hour

- Hourly Demand for water = $(81000/24)$ Litres/Hour = 3375 Litres/Hour

Overhead Water Tank Capacity & Design:

Designing the Storage Capacity as per Table 16 of SP 35

Storage Capacity as per clause 5.4.2.3 of SP 35

- Storage = Population * 70L = 300 * 70L = 21,000 Litres
- Maximum Storage = 0.5 * Daily Supply = 0.5 * 81000 = 40500 Litres

Therefore, We will take storage of Overhead water tank as 40500 Litres

Let's consider dimensions of Storage tank as: Radius = 2m, Height = 1m

Volume of a Single Storage Tank = $3.14 * 2 * 2 * 1 * 1000 = 12566$ Litres

Number of Tanks required = $40500/12566 = 3.223$

Since the number of tanks is a decimal number, let's install 4 storage tanks of Radius = 2m, Height = 1m, each having a capacity of 12500 Litres.

Therefore, the total installed water tank storage capacity = 50,000 L

Design of sewerage pipe system:

A two pipe system will be provided in the residential blocks for disposal of sewage where soil pipes are connected to the building drain directly and the waste pipes are connected to the building drain using a trapped gully. A main ventilation pipe (MVP) is provided for the main waste pipes (MWP) and the main soil pipe (MSP)

1. Waste Appliances (Internal diameters) (Table 54 of SP 35)

Waste Appliances	Internal Diameter
Wash Basin	30 mm
Bathroom	40 mm
Water Closet	50 mm

2. Horizontal branches for each floor(Diameters) (Table 52 of SP 35)

Floor	Blocks	Blocks
	MWP	MSP
All floors	100 mm	100 mm

(MWP and MSP columns indicate the diameter of the horizontal branches joining MWP and MSP respectively)

MWP and MSP (Table 53 of SP) Diameter provided for all the MWP and MSP is 100mm. MVP (Clause 6.7.5.3 of SP) The diameter provided for the MVP is 50mm

Design of sewer pipe:

- Total water demand per day = 81000 Litres/day
- Now assume 90% of total consumption to reach sewer
Here, peak factor = 3 (since population < 20,000)
- Total capacity = $81000 * 0.9 * 3$ Litres/day = 2531.25 cm³/s

- Using Manning's formula $Q = (AR^{(2/3)}S^{(1/2)})/n$

Let's consider slope of main sewer pipe = 1/1000, Manning's coefficient = 0.012

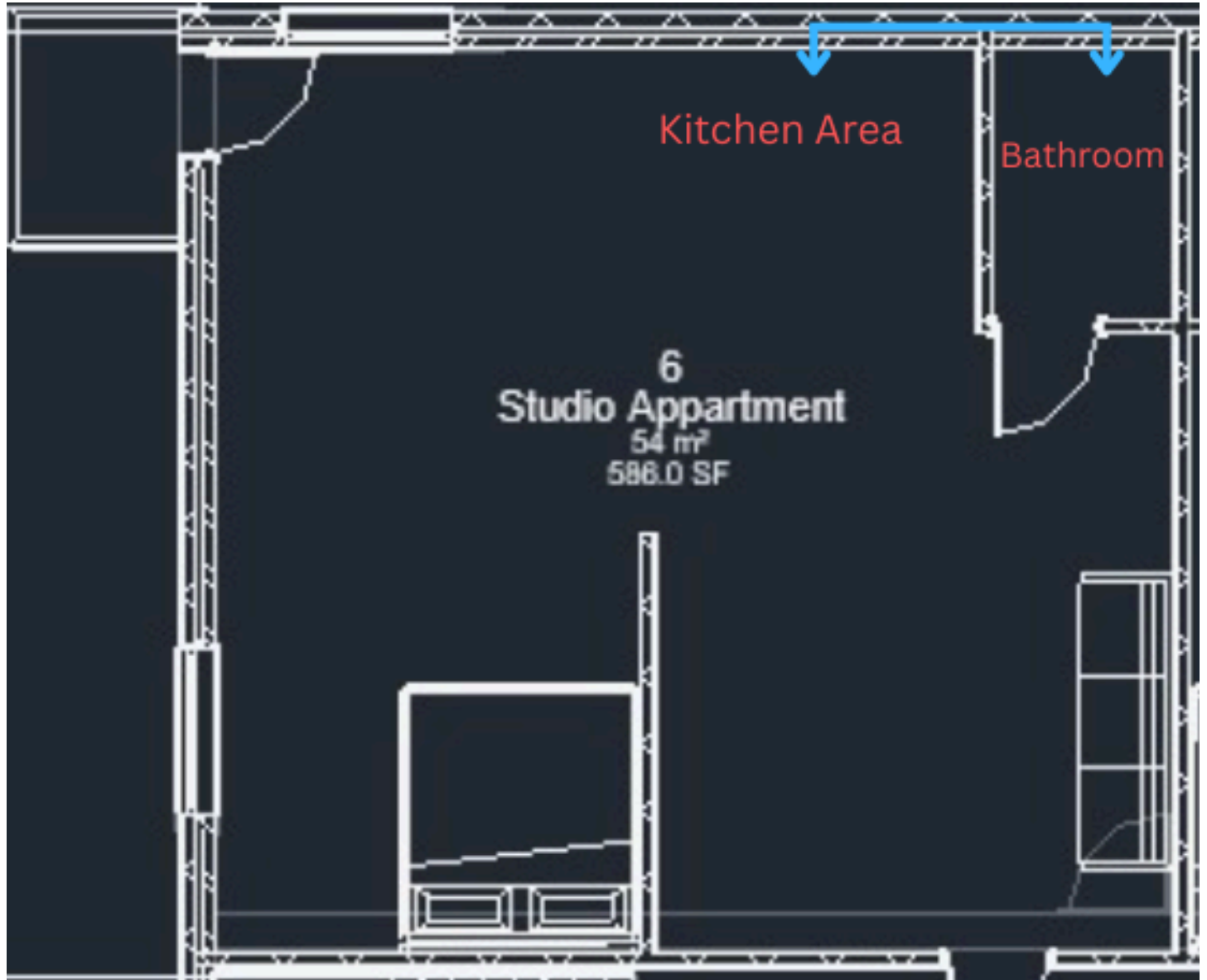
$$\Rightarrow 2531.25 = 1/0.012 * ((\pi * d^2)/4) * ((d/4)^{(2/3)}) * (1/1000)^{(1/2)}$$

$$\Rightarrow d = 20.337 \text{ cm}$$

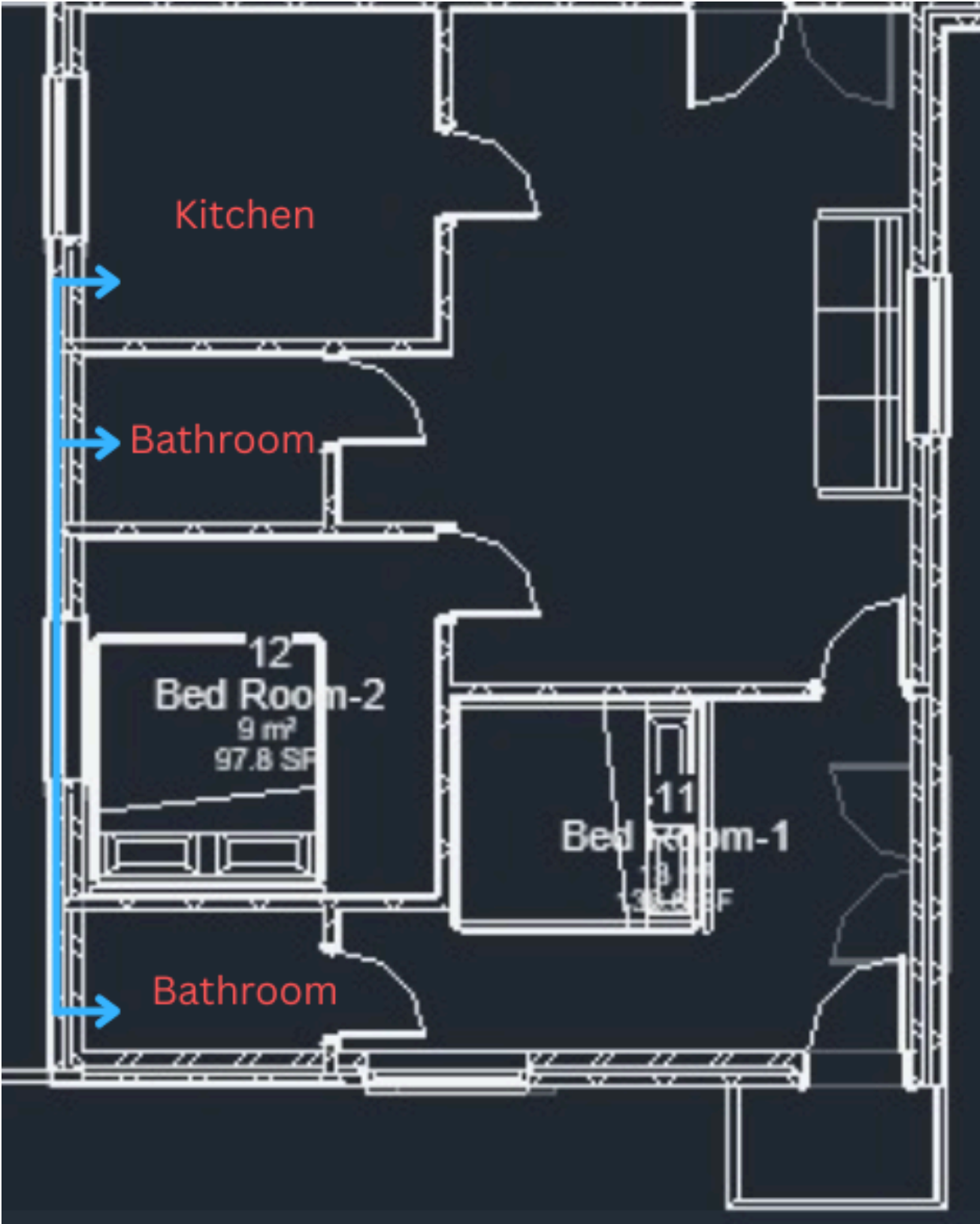
=> Provide 210 mm diameter pipe for the sewer pipe connecting the sewage from all flats to the municipal sewage system

Plumbing Layout Plan:

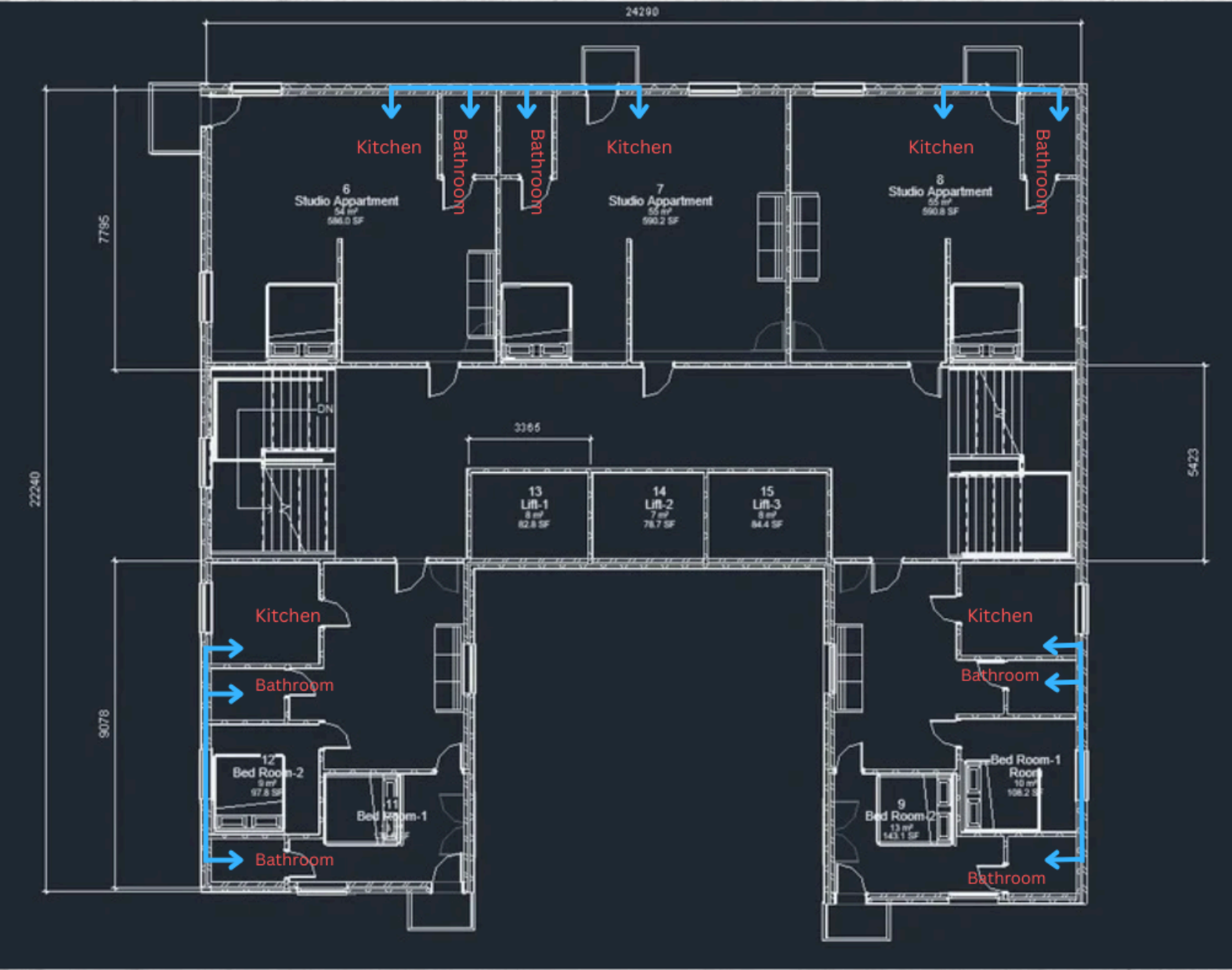
Studio Apartment:



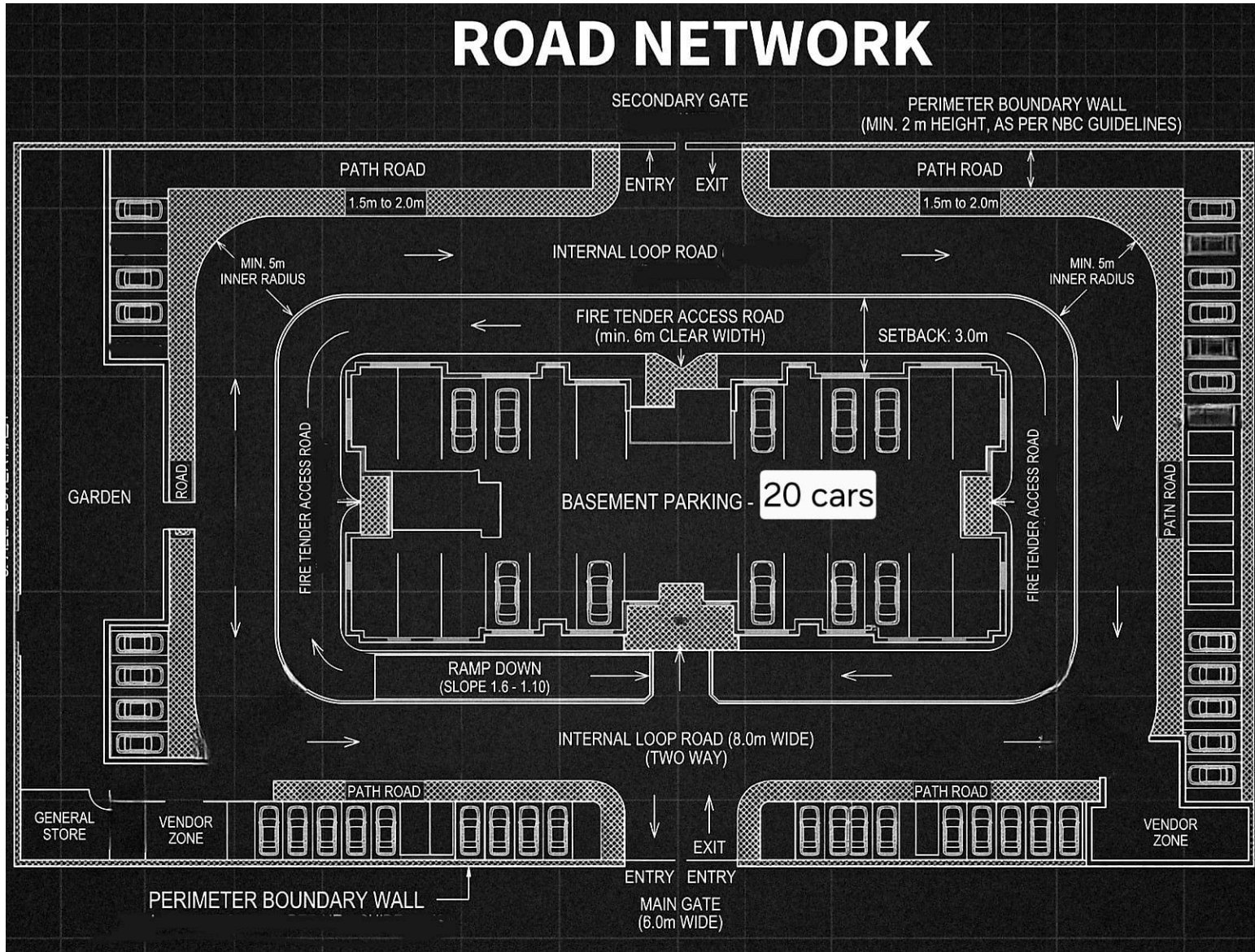
2-BHK Apartment:



Floor Plumbing Layout:



Road Network:



- Since the area of the entire building premises is around **467 sq.m**, it's reasonable to assume that there's a road on all **4 sides** of the building.
- Ramp is provided for parking inside the building in the basement (20 parking plots) and parking area for remaining apartments is provided around the building at ground floor level.
- 6 m width allows two way traffic and is sufficient for cars and service vehicles movement

Cost Estimation:

The estimation is prepared following Indian construction practices using CPWD Schedule of Rates (Chandigarh basis) and standard engineering assumptions where necessary.

Floor Details:

Built-up Area per Floor: 467 m²

Total Built-up Area: 7472.7 m²

Basement & Ground Height: 2.5 m

Typical Floors Height: 3.0 m

Structural System:

Structural Type: RCC Frame with Shear Wall Core

Slab Thickness: 120 mm

Beam Size: 400 mm × 350 mm

Column Size: 700 mm × 500 mm

Shear Wall Thickness: 200 mm

Materials:

Concrete: M30

Reinforcement Steel: Fe500

Brick: Fly ash bricks

Finishes:

Flooring: Tiles

Plaster: Cement plaster (internal & external)

Paint: Emulsion

Doors: Wooden, Windows: uPVC

Services:

Plumbing: Included

Electrical: Included

Lifts: 3

Basement: Yes

Foundation:

Foundation Type: Isolated Footings

Total Unique Footings: 44

Length/Width: 4.2 m to 8.0 m

Depth: 0.65 m to 1.10 m

Assumptions:

Structural Quantities:

Each footing type occurs once (based on column grid interpretation)

Number of columns $\approx$ 40–45

Beam volume approximated as $0.3 \text{ m}^3/\text{m}^2$

Steel consumption:

Foundation: $120 \text{ kg}/\text{m}^3$

Superstructure: $110 \text{ kg}/\text{m}^3$

Architectural Quantities:

Brickwork: 0.3 m^3 per m^2 built-up area

Plaster Area: $2.5 \times$ built-up area

Doors: 1 per 25 m^2

Windows: $1.5 \times$ number of doors

Shear Wall:

Located in core region only, Occupies approximately 15% of floor area

Rates:

Adopted from CPWD Schedule of Rates (Chandigarh, latest available basis): -

RCC: ₹8000/m³

Steel: ₹75/ kg

Brickwork: ₹6000/m³

Plaster: ₹250/m²

Flooring: ₹1200/m²

Additional Costs:

Services: 20% of civil cost

Lift Cost: ₹50 lakh per lift

External Works: Lump sum ₹80 lakh

Contractor Profit: Included in rates

Contingency: Included in rounding

QUANTITY TAKEOFF:

RCC Works:

Component	Quantity (m ³)
Foundation	1258
Slab	781
Beams	1954
Columns	700
Shear Walls	244
Total RCC	4937 m³

Reinforcement Steel:

Component	Quantity
Foundation Steel	151 MT
Superstructure Steel	392 MT
Total Steel	543 MT

Masonry:

Item	Quantity
Brickwork	1954 m ³

Finishes:

Item	Quantity
Plaster	16281 m ²
Flooring	6512.7 m ²

Openings:

Item	Quantity
Doors	260 Nos
Windows	390 Nos

COST ESTIMATION (CPWD BASIS):

Civil Cost:

Item	Amount
RCC	₹3.95 Cr
Steel	₹4.07 Cr
Brickwork	₹1.17 Cr
Plaster	₹0.41 Cr
Flooring	₹0.78 Cr
Subtotal (Civil)	₹10.4 Cr

Additional Costs:

Item	Amount
Services (20%)	₹2.08 Cr
Lifts (3 Nos)	₹1.50 Cr
External Works	₹0.80 Cr

TOTAL PROJECT COST:

Total Estimated Cost: ₹14.8 Crore (Approx.)

Cost per m² = ₹19,800/m²

ANALYTICAL INSIGHTS:

Cost Distribution:

Steel: ~40%

RCC: ~27%

Finishes: ~12%

Services: ~14%

Key Observations:

Steel is the dominant cost component due to high-rise structure.

Foundation cost is significant due to large footing sizes.

Shear wall contributes to stability but increases RCC volume.

CONCLUSION:

The estimated project cost of ₹14.8 Crore aligns with typical high-rise residential construction costs in India (₹18,000–₹25,000 per m²).

Thank You